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March 7, 2023

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Attn: Mr. Yoichi Shimokobe, CEO

Subject: Roof Tech RT-Mini II Mount – Structural Analysis

SML Job No.: 471-22

Dear Mr. Shimokobe:

We have analyzed the Roof Tech RT-Mini II photovoltaic (PV) panel roof mount system and determined that, for the configurations and criteria described below, it is in compliance with the applicable sections of the following Reference Documents:

Codes: ASCE/SEI 7-16 Min. Design Loads for Buildings & Other Structures
International Building Code 2018 & 2021 Editions
Other: Aluminum Design Manual, 2015 & 2020 Editions

The Roof Tech RT-Mini II photovoltaic (PV) panel roof mount system is comprised of a 6000 series aluminum RT-Mini II base. An appropriately load rated 'L-Foot', by others, may be attached to the RT-Mini II base with a stainless steel SS304 5/16" diameter flange bolt or 5/16" diameter bolt with 5/8" OD washer, and flange nut. An appropriately load rated rail, by others, may be attached to the 'L-Foot' per the rail manufacturer's installation instructions. The installation of the RT-Mini II must be with the long direction parallel to the roof framing and in accordance with Roof Tech's Installation Manual. The system is attached to the roof wood substrate with either SS304 5.0 mm x 60 mm or SS304 5.0 mm x 90 mm wood screws.

Three wood substrates were tested with the fasteners also passing through two layers of composite asphalt roof shingles and 2 layers of building paper. The three (3) tested wood substrate consisted of 15/32" thick plywood sheathing over a 2x4 DF #2 rafter (2x truss top chord OK by inspection), 7/16" thick OSB only, and 15/32" thick plywood only, See Exhibits 'A1' & 'A2'. **Two (60mm or 90mm long) wood screws are required at the rafter installations, while five (60mm long) wood screws are required at the rafter offset, OSB only, and plywood only installations.**

The testing data provided by Institute of Building Technology (IBT) report no.: 2426-21007-002 project no.: 34578 dated April 19, 2021, shows that the critical failure occurs in one of two ways; pullout of the wood screws from the 2x rafter and/or yielding of the attached components in

shear. Maximum allowable values for these failure conditions were derived using a safety factor 3.0 based on the average value. The testing was performed in general accordance with IAPMO Uniform Evaluation Services *Evaluation Criteria for Joist Hangers and Miscellaneous Connectors*, EC002-2018 and ASTM D1761-2020, *Standard Test Methods for Mechanical Fasteners in Wood*.

Compression load capacity on the OSB and plywood installations were determined by testing conducted at Western Technologies Inc. (WT). The data is represented in job no. 2163XD260 event no. G260-3 (OSB) dated January 3, 2014 and job no. 2163XD260 event no. G260-6 (Plywood) dated May 30, 2014. The safety factor for downward compression on the sheathing is 2.0 based on the average value.

The results of the above testing are presented below:

The maximum allowable uplift, down force, and shear force for 7/16" thick OSB only is 113.7 lb, 258.0 lb, and 216.7 lb respectively. The maximum allowable uplift, down force, and shear force for 15/32" thick plywood only is 138.0 lb, 556.0 lb, and 228.8 lb respectively. The maximum allowable uplift and shear force for 15/32" thick plywood sheathing over a 2x4 DF-L #2 rafter using (2) 60mm long screws is 569.9 lb and 404.4 lb respectively. The maximum allowable uplift and shear force for either 7/16" thick OSB or 15/32" thick plywood sheathing over a 2x4 DF-L #2 rafter in the offset position, see Exhibit 'A2', using (5) 60mm long screws is 325.7 lb and 411.0 lb respectively. The maximum allowable uplift and shear force for 15/32" thick plywood sheathing over a 2x4 DF-L #2 rafter using (2) 90mm long screws is 613.2 lb and 469.9 lb respectively.

The attached tables below show the maximum spacing of the RT-Mini II for the combined loading criteria of wind speed, roof slope, and snow loads which produce reaction loads less than the allowable loads listed above. The effect of seismic loads (for all design categories A-E) have been determined to be less than the effect due to wind loads in all load conditions and combinations. The assumed topography is flat with a $K_{zt} = 1.0$. Therefore, the maximum allowable spacing for common load cases due to dead, wind and snow loads are the controlling load cases.

The maximum panel height is measured from the top of the roof to the top of the panel, and shall be 7" for rafter installations, and 5" for OSB and Plywood Only installations, see Exhibits 'A1' & 'A2', for the allowable PV panel layouts, and 'L-Foot' and rail configurations, see Exhibit 'A3'. The maximum rail cantilever shall be limited to the smaller of 40% of the maximum mount spacing in the tables below and the rail manufacturer's specified maximum cantilever. It is the responsibility of the contractor to verify that the span rating of the 'L-Foot' and rails that are used for the given project and use the lower of those spans or the mount spacing given in these tables. RT-Mini II mounts shall be skip loaded on the roof where possible, see Exhibits 'A4' & 'A5', and provide thermal expansion splices along the rail at intervals not to exceed manufacturer's recommendations or 14 feet o.c. (refer to the rail manufacturer's thermal expansion splice detailing for proper installation).

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* 30 ft Mean Roof Height

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The following conversion chart has been created to help clarify the ASCE 7-16 roof zones, which vary from gable to hip and for various roof pitches, while also keeping the spacing tables in as simple a form as possible.

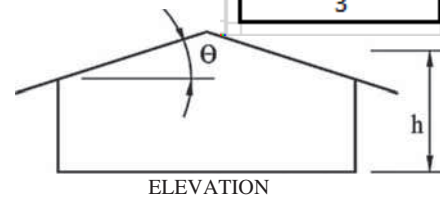
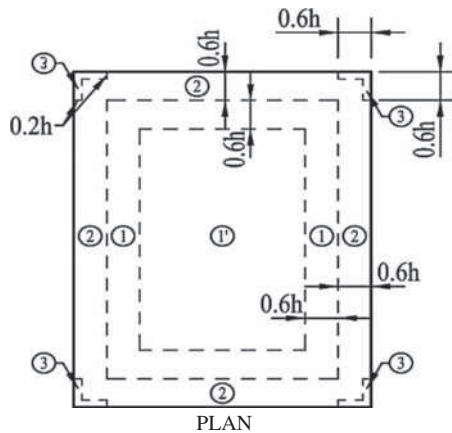
Each ASCE 7-16 figure 30.3-2A to 30.3-2H (See pages 6 to 12) has up to six (6) roof zones, which we have converted to three (3) “Roof Areas” based on their common GCp values.

Conversion Chart - SML Table Roof Areas to ASCE 7-16 Roof Zones							
Roof Type	Roof Slopes, (deg)	Roof Areas Per SML Tables	Roof Zones Per ASCE 7-16	Effective Wind Area and GCp Range for Leeward Wind, (ft ²)		Effective Wind Area and GCp Range for Windward Wind, (ft ²)	
Gable Roof	0 to 7	1	1', 1	10 to 500	-1.7 to -1	10 to 100	0.3 to 0.2
		2	2	10 to 500	-2.3 to -1.4	10 to 100	0.3 to 0.2
		3	3	10 to 500	-3.2 to -1.4	10 to 100	0.3 to 0.2
	7 to 20	1	1, 2e	20 to 100	-2 to -0.5	2 to 100	0.7 to 0.3
		2	2n, 2r, 3e	10 to 250	-3 to -1	2 to 100	0.7 to 0.3
		3	3r	10 to 100	-3.6 to -1.8	2 to 100	0.7 to 0.3
	20 to 27	1	1, 2e	20 to 300	-1.5 to -0.8	2 to 100	0.7 to 0.3
		2	2n, 2r, 3e	10 to 150	-2.5 to -1.2	2 to 100	0.7 to 0.3
		3	3r	4 to 50	-3.6 to -1.8	2 to 100	0.7 to 0.3
	27 to 45	1	1, 2e, 2r	10 to 100	-1.8 to -0.8	10 to 100	0.9 to 0.5
		2	2n, 3r	10 to 200	-2 to -1	10 to 100	0.9 to 0.5
		3	3e	2 to 300	-3.2 to -1	10 to 100	0.9 to 0.5
Hip Roof	0 to 7	NA	NA	NA	NA	NA	NA
	7 to 20	1	1	20 to 100	-1.8 to -1	10 to 100	0.7 to 0.3
		2	2r	10 to 200	-2.4 to -1.3	10 to 100	0.7 to 0.3
		3	2e, 3	10 to 200	-2.6 to -1.4	10 to 100	0.7 to 0.3
	20 to 27	1	1	10 to 100	-1.4 to -0.8	10 to 100	0.7 to 0.3
		2	2e, 2r, 3	10 to 200	-2 to -1	10 to 100	0.7 to 0.3
		3	NA	NA	NA	NA	NA
	27 to 45	1	1	10 to 200	-1.5 to -0.7	3 to 100	0.9 to 0.3
		2	2e, 2r	7 to 100	-2.8 to -1	3 to 100	0.9 to 0.3
		3	3	5 to 50	-3.6 to -1	3 to 100	0.9 to 0.3

Combining Rafter and Deck-Only Mounts:

It is permissible to combine rafter and deck-only mounting as shown at the end of this letter in Exhibits ‘B1’ & ‘B2’. In summary, the rafter-to-deck spacing shall be the average of the rafter spacing and deck-only (OSB or Plywood Only) spacing. When using combined rafter and deck-only mounting the maximum panel height shall be either 5” with an average PV panel gap of 0.5” or 7” with an average PV panel gap of 0.7”, see table note 7 for more information.

Diagrams



Roof Areas Per SML Tables	Roof Zones Per ASCE 7-16
1	1', 1
2	2
3	3

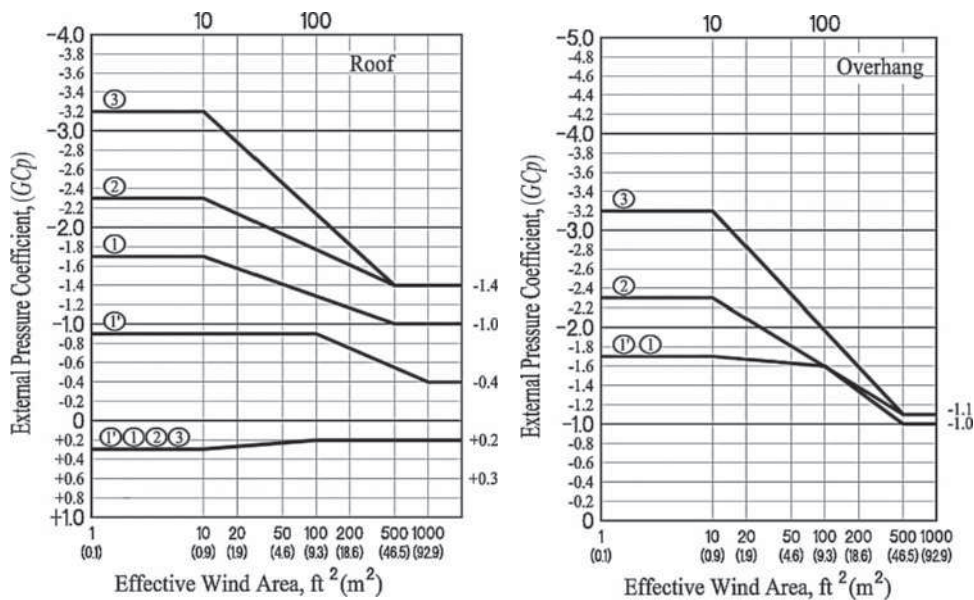
Notation

B = Horizontal dimension of building measured normal to wind direction, in ft (m).

h = Eave height shall be used for $\theta = 10^\circ$.

θ = Angle of plane of roof from horizontal, in degrees.

External Pressure Coefficients

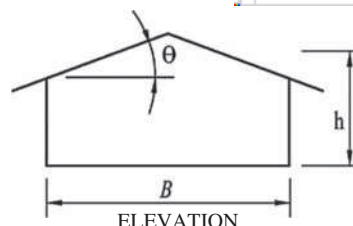
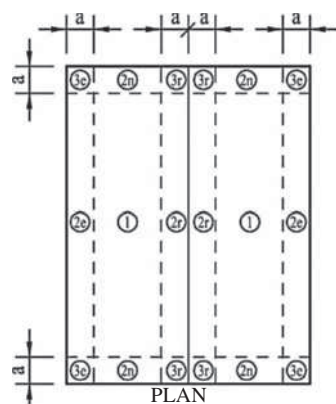


Notes

1. Vertical scale denotes (GC_p) to be used with q_h .
2. Horizontal scale denotes effective wind area, in ft^2 (m^2).
3. Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
4. Each component shall be designed for maximum positive and negative pressures.
5. If a parapet equal to or higher than 3 ft (0.9 m) is provided around the perimeter of the roof with $\theta \leq 7^\circ$, the negative values of (GC_p) in Zone 3 shall be equal to those for Zone 2, and positive values of (GC_p) in Zones 2 and 3 shall be set equal to those for wall Zones 4 and 5, respectively, in Fig. 30.3-1.
6. Values of (GC_p) for roof overhangs include pressure contributions from both upper and lower surfaces.
7. If overhangs exist, the lesser horizontal dimension of the building shall not include any overhang dimension, but the edge distance, a , shall be measured from the outside edge of the overhang.

FIGURE 30.3-2A Components and Cladding [$h \leq 60$ ft ($h \leq 18.3$ m)]: External Pressure Coefficients, (GC_p) , for Enclosed and Partially Enclosed Buildings—Gable Roofs, $\theta \leq 7^\circ$

Diagrams

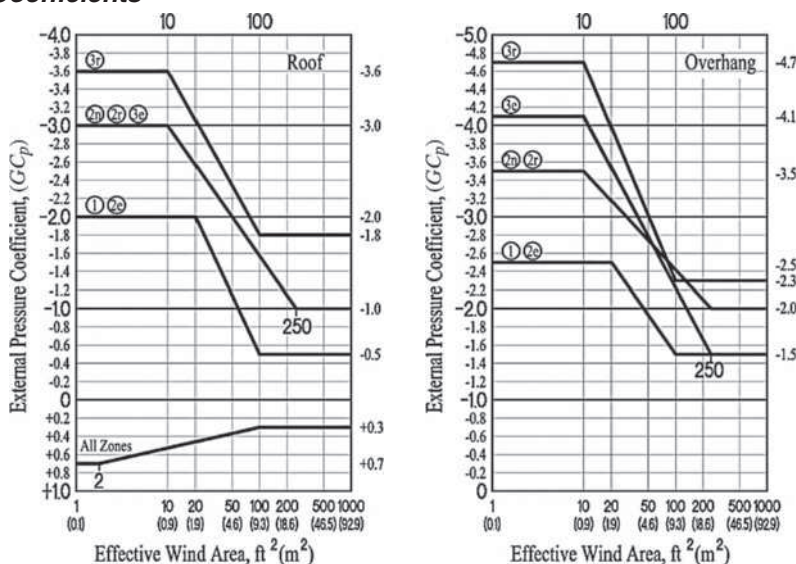


Roof Areas Per SML Tables	Roof Zones Per ASCE 7-16
1	1, 2e
2	2n, 2r, 3e
3	3r

Notation

a = 10% of least horizontal dimension or $0.4h$, whichever is smaller, but not less than either 4% of least horizontal dimension or 3 ft (0.9 m). If an overhang exists, the edge distance shall be measured from the outside edge of the overhang. The horizontal dimensions used to compute the edge distance shall not include any overhang distances.
 B = Horizontal dimension of building measured normal to wind direction, in ft (m).
 h = Mean roof height, in ft (m), except that eave height shall be used for $\theta \leq 10^\circ$.
 θ = Angle of plane of roof from horizontal, in degrees.

External Pressure Coefficients

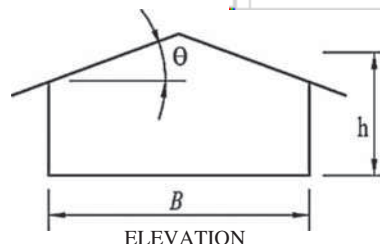
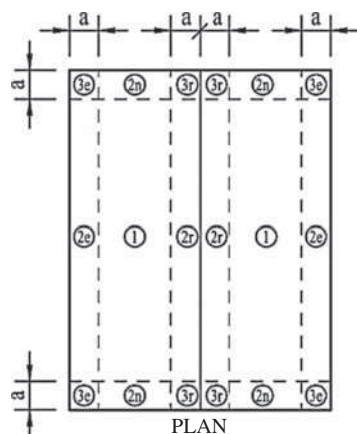


Notes

1. Vertical scale denotes (GC_p) to be used with q_h .
2. Horizontal scale denotes effective wind area, in ft^2 (m^2).
3. Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
4. Each component shall be designed for maximum positive and negative pressures.
5. Values of (GC_p) for roof overhangs include pressure contributions from both upper and lower surfaces.
6. If overhangs exist, the lesser horizontal dimension of the building shall not include any overhang dimension, but the edge distance, a , shall be measured from the outside edge of the overhang.

FIGURE 30.3-2B Components and Cladding [$h \leq 60$ ft ($h \leq 18.3$ m)]: External Pressure Coefficients, (GC_p) , for Enclosed and Partially Enclosed Buildings—Gable Roofs, $7^\circ < \theta \leq 20^\circ$

Diagrams



Roof Areas Per SML Tables	Roof Zones Per ASCE 7-16
1	1, 2e
2	2n, 2r, 3e
3	3r

Notation

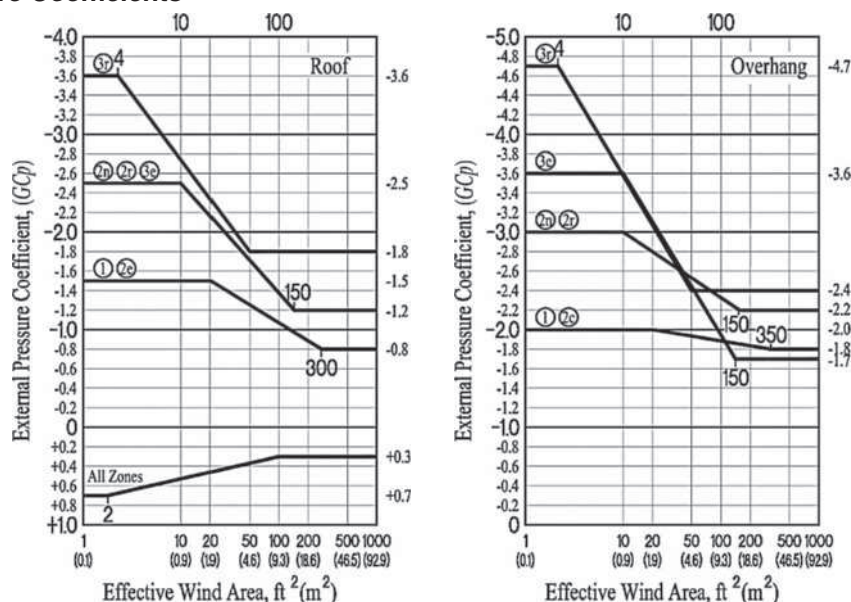
a = 10% of least horizontal dimension or $0.4h$, whichever is smaller, but not less than either 4% of least horizontal dimension or 3 ft (0.9 m). If an overhang exists, the edge distance shall be measured from the outside edge of the overhang. The horizontal dimensions used to compute the edge distance shall not include any overhang distances.

B = Horizontal dimension of building measured normal to wind direction, in ft (m).

h = Mean roof height, in ft (m).

θ = Angle of plane of roof from horizontal, in degrees.

External Pressure Coefficients

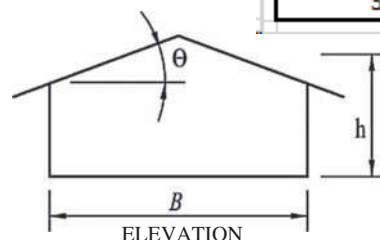
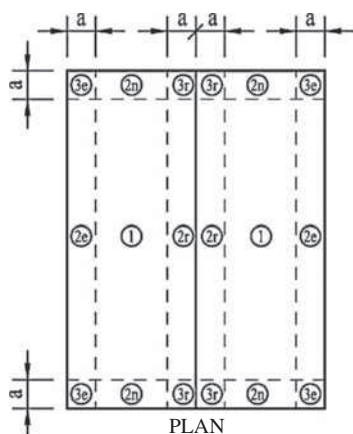


Notes

1. Vertical scale denotes (GCP) to be used with q_h .
2. Horizontal scale denotes effective wind area, in $ft^2(m^2)$.
3. Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
4. Each component shall be designed for maximum positive and negative pressures.
5. Values of (GCP) for roof overhangs include pressure contributions from both upper and lower surfaces.
6. If overhangs exist, the lesser horizontal dimension of the building shall not include any overhang dimension, but the edge distance, a , shall be measured from the outside edge of the overhang.

FIGURE 30.3-2C Components and Cladding [$h \leq 60$ ft ($h \leq 18.3$ m)]: External Pressure Coefficients, (GCP) , for Enclosed and Partially Enclosed Buildings—Gable Roofs, $20^\circ < \theta \leq 27^\circ$

Diagrams



Roof Areas Per SML Tables	Roof Zones Per ASCE 7-16
1	1, 2e, 2r
2	2n, 3r
3	3e

Notation

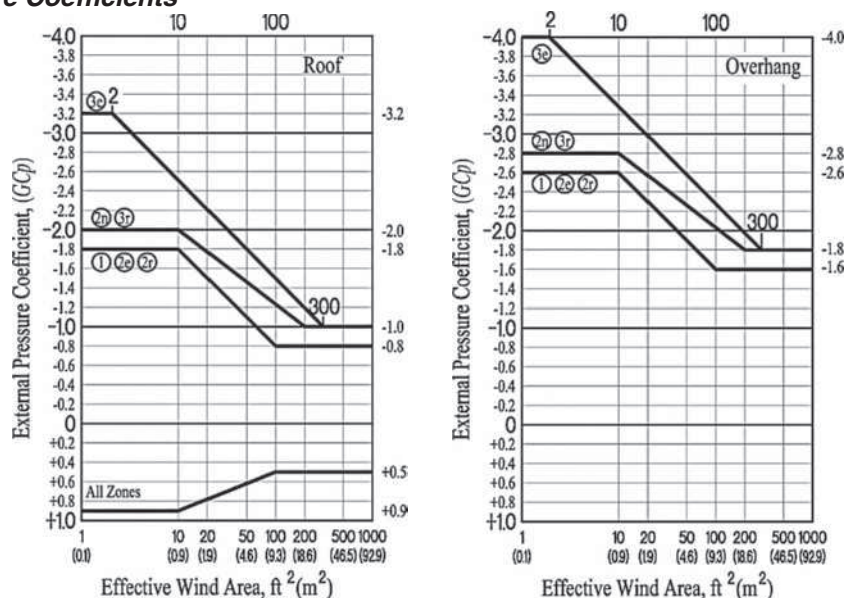
a = 10% of least horizontal dimension or $0.4h$, whichever is smaller, but not less than either 4% of least horizontal dimension or 3 ft (0.9 m). If an overhang exists, the edge distance shall be measured from the outside edge of the overhang. The horizontal dimensions used to compute the edge distance shall not include any overhang distances.

B = Horizontal dimension of building measured normal to wind direction, in ft (m).

h = Mean roof height, in ft (m).

θ = Angle of plane of roof from horizontal, in degrees.

External Pressure Coefficients

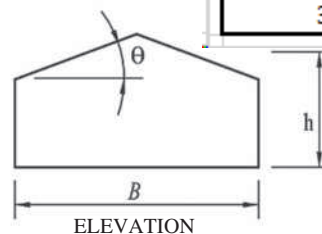
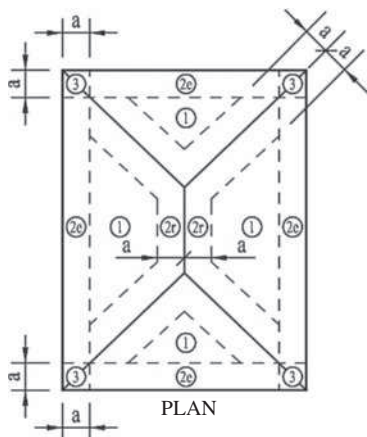


Notes

1. Vertical scale denotes (GCP) to be used with q_h .
2. Horizontal scale denotes effective wind area, in ft^2 (m^2).
3. Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
4. Each component shall be designed for maximum positive and negative pressures.
5. Values of (GCP) for roof overhangs include pressure contributions from both upper and lower surfaces.
6. If overhangs exist, the lesser horizontal dimension of the building shall not include any overhang dimension, but the edge distance, a , shall be measured from the outside edge of the overhang.

FIGURE 30.3-2D Components and Cladding [$h \leq 60$ ft ($h \leq 18.3$ m)]: External Pressure Coefficients, (GCP), for Enclosed and Partially Enclosed Buildings—Gable Roofs, $27^\circ < \theta \leq 45^\circ$

Diagrams



Roof Areas Per SML Tables	Roof Zones Per ASCE 7-16
1	1
2	2r
3	2e, 3

Notation

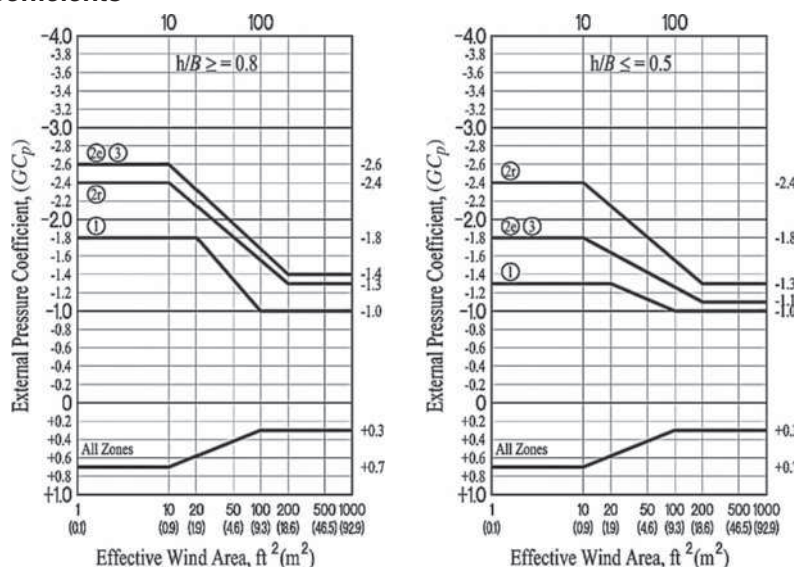
a = 10% of least horizontal dimension or $0.4h$, whichever is smaller, but not less than either 4% of least horizontal dimension or 3 ft (0.9 m). If an overhang exists, the edge distance shall be measured from the outside edge of the overhang. The horizontal dimensions used to compute the edge distance shall not include any overhang distances.

h = Mean roof height, in ft (m), except that eave height shall be used for $\theta \leq 10^\circ$.

B = Horizontal dimension of building measured normal to wind direction, in ft (m).

θ = Angle of plane of roof from horizontal, in degrees.

External Pressure Coefficients

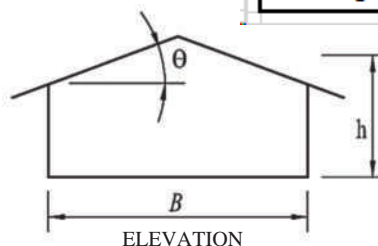
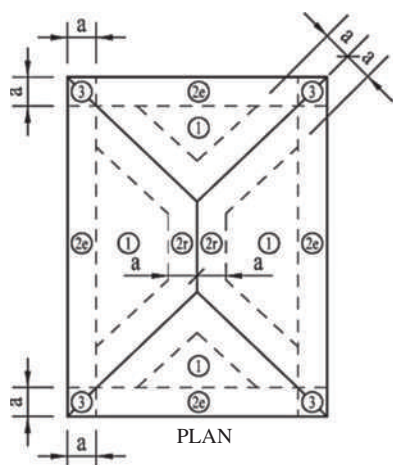


Notes

1. Vertical scale denotes (G_{CP}) to be used with q_h .
2. Horizontal scale denotes effective wind area, in $\text{ft}^2 (\text{m}^2)$.
3. Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
4. Each component shall be designed for maximum positive and negative pressures.
5. If overhangs exist, the lesser horizontal dimension of the building shall not include any overhang dimension, but the edge distance, a , shall be measured from the outside edge of the overhang.
6. Interpolation of (G_{CP}) between the two different h/B values is required for $0.5 < h/B < 0.8$.
7. B for Zone 3 is the least horizontal dimension. B for Zones 1 and 2e is normal to the building width and normal to the eave defining Zone 2e.

FIGURE 30.3-2E Components and Cladding [$h \leq 60$ ft ($h \leq 18.3$ m)]: External Pressure Coefficients, (G_{CP}), for Enclosed and Partially Enclosed Buildings—Hip Roofs, $7^\circ < \theta \leq 20^\circ$ (Roof)

Diagrams

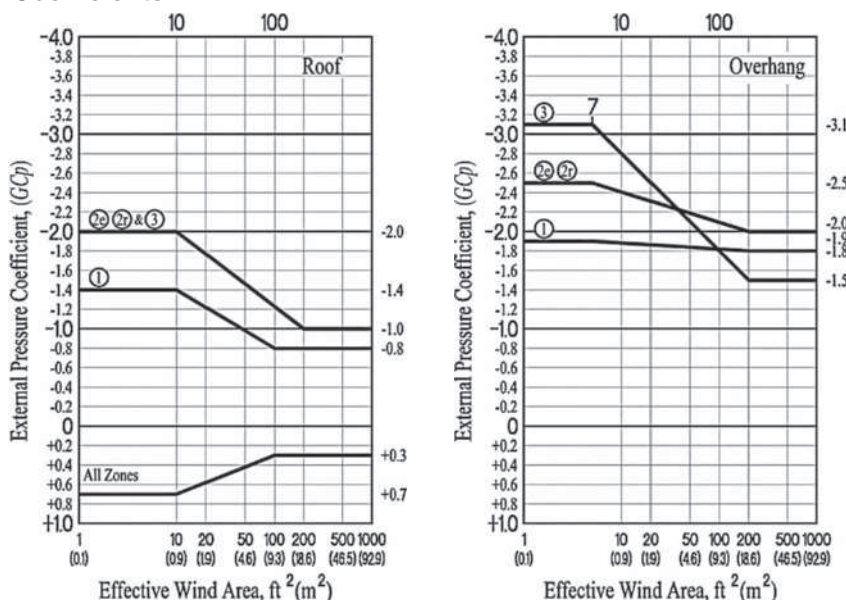


Roof Areas Per SML Tables	Roof Zones Per ASCE 7-16
1	1
2	2e, 2r, 3
3	NA

Notation

a = 10% of least horizontal dimension or $0.4h$, whichever is smaller, but not less than either 4% of least horizontal dimension or 3 ft (0.9 m). If an overhang exists, the edge distance shall be measured from the outside edge of the overhang. The horizontal dimensions used to compute the edge distance shall not include any overhang distances.
 B = Horizontal dimension of building measured normal to wind direction, in ft (m).
 h = Mean roof height, in ft (m).
 θ = Angle of plane of roof from horizontal, in degrees.

External Pressure Coefficients

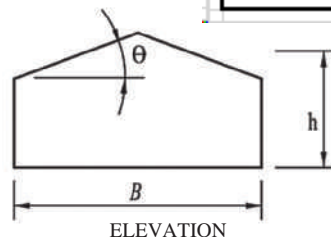
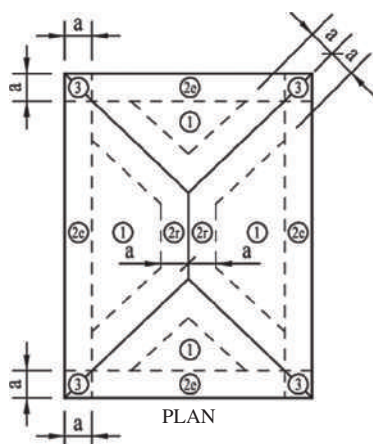


Notes

1. Vertical scale denotes (GCP) to be used with q_h .
2. Horizontal scale denotes effective wind area, in ft^2 (m^2).
3. Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
4. Each component shall be designed for maximum positive and negative pressures.
5. Values of (GCP) for roof overhangs include pressure contributions from both upper and lower surfaces.
6. If overhangs exist, the lesser horizontal dimension of the building shall not include any overhang dimension, but the edge distance, a , shall be measured from the outside edge of the overhang.

FIGURE 30.3-2G Components and Cladding [$h \leq 60$ ft ($h \leq 18.3$ m)]: External Pressure Coefficients, (GCP), for Enclosed and Partially Enclosed Buildings—Hip Roofs, $20^\circ < \theta \leq 27^\circ$ (Roof and Overhang)

Diagrams



Roof Areas Per SML Tables	Roof Zones Per ASCE 7-16
1	1
2	2e, 2r
3	3

Notation

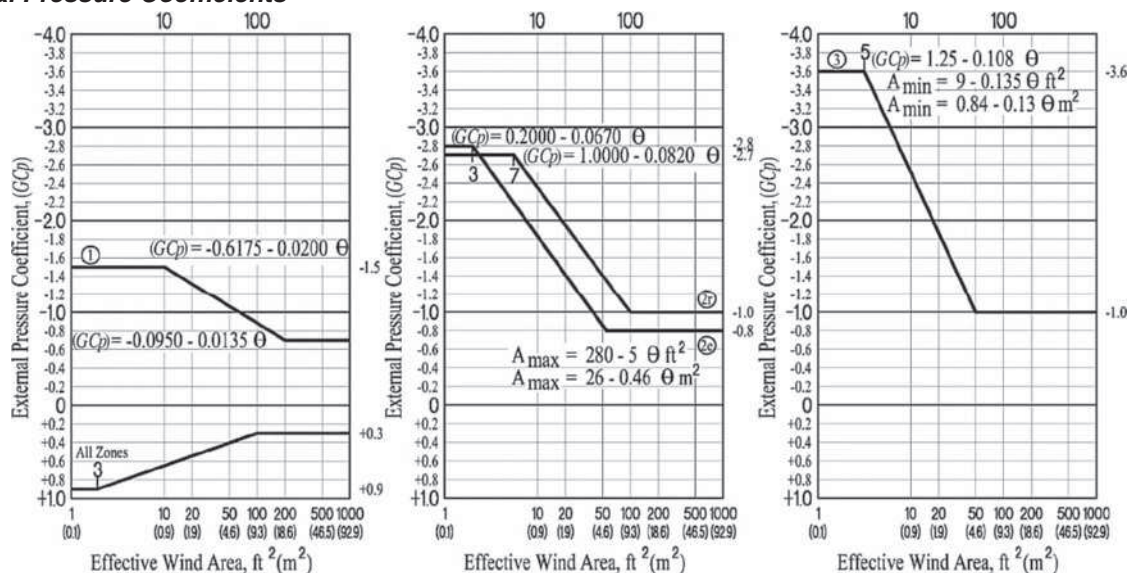
a = 10% of least horizontal dimension or $0.4h$, whichever is smaller, but not less than either 4% of least horizontal dimension or 3 ft (0.9 m). If an overhang exists, the edge distance shall be measured from the outside edge of the overhang. The horizontal dimensions used to compute the edge distance shall not include any overhang distances.

B = Horizontal dimension of building measured normal to wind direction, in ft (m).

h = Mean roof height, in ft (m).

θ = Angle of plane of roof from horizontal, in degrees.

External Pressure Coefficients



Notes

- Vertical scale denotes (G_{Cp}) to be used with q_h .
- Horizontal scale denotes effective wind area, in ft^2 (m^2).
- Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
- Each component shall be designed for maximum positive and negative pressures.
- If overhangs exist, the lesser horizontal dimension of the building shall not include any overhang dimension, but the edge distance, a , shall be measured from the outside edge of the overhang.
- $A_{\min}$ = the minimum tributary area (i.e., areas less than $A_{\min}$ are to use (G_{Cp}) value for $A_{\min}$).
- $A_{\max}$ = the maximum tributary area (i.e., areas greater than $A_{\max}$ are to use (G_{Cp}) value for $A_{\max}$).
- (G_{Cp}) values given for roof slope, $\theta = 45^\circ$; for other slopes use the equations.

FIGURE 30.3-2H Components and Cladding [$h \leq 60$ ft ($h \leq 18.3$ m)]: External Pressure Coefficients, (G_{Cp}) , for Enclosed and Partially Enclosed Buildings—Hip Roofs, $27^\circ < \theta \leq 45^\circ$ (Roof)

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

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Roof Tech - Mini Gen II - ASCE 7-16 Gable - Flush-Mount - 15 Foot Max Mean Roof Height - Plywood Only - Portrait 60 Cell - 60 mm Long Screws																																								
Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->	95			100			105			110			115			120			130			140			150			160			170			180		
				Roof Area ---->				1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3			
		0	0.0	Non-Exposed / *Exposed	37	37/*36	37/*25	37	37/*32	35/*22	37	37/*28	31/*20	37/*36	37/*26	28/*18	37/*33	37/*23	25/*16	37/*30	33/*21	23/*15	37/*25	28/*18	19/*12	33/*21	23/*15	16/*10	28/*18	20/*13	14/*9	24/*16	17/*11	12/*8	21/*14	15/*10	11/*7	19/*12	13/*9	9/*6
		1	4.8	Non-Exposed / *Exposed	31	31	31/*24	31	31	31/*22	31	31/*28	31/*20	31	31/*25	28/*18	31	31/*23	25/*16	31/*29	31/*21	23/*15	31/*25	28/*18	19/*12	31/*21	23/*15	16/*10	28/*18	20/*13	14/*9	24/*16	17/*11	12/*8	21/*14	15/*10	11/*7	19/*12	13/*9	9/*6
		2	9.5	Non-Exposed / *Exposed	27	27/*26	27/*21	27	27/*23	27/*19	27	27/*21	27/*17	27	27/*19	24/*15	27	27/*17	22/*14	27/*24	24/*16	20/*13	27/*20	20/*13	17/*11	27/*17	17/*11	14/*9	23/*15	15/*10	12/*8	20/*13	13/*8	11/*7	18/*11	11/*7	9/*6	16/*10	10/*6	8/*5
		3	14.0	Non-Exposed / *Exposed	24	24	24/*21	24	24/*23	24/*19	24	24/*21	24/*17	24	24/*19	24/*15	24	24/*17	22/*14	24	24/*16	20/*13	24/*20	20/*13	17/*11	24/*17	17/*11	14/*9	23/*15	15/*10	12/*8	20/*13	13/*8	11/*7	18/*11	11/*7	9/*6	16/*10	10/*6	8/*5
		4	18.4	Non-Exposed / *Exposed	24	24	24/*21	24	24/*23	24/*19	24	24/*21	24/*17	24	24/*19	24/*15	24	24/*17	22/*14	24	24/*15	20/*13	24/*20	20/*13	17/*11	24/*17	17/*11	14/*9	23/*15	15/*10	12/*8	20/*13	13/*8	11/*7	17/*11	11/*7	9/*6	15/*10	10/*6	8/*5
		5	22.6	Non-Exposed / *Exposed	24	24	24/*22	24	24/*24	24/*19	24	24	24/*17	24	24/*23	24/*15	24	24/*21	23/*14	24	24/*19	20/*13	24	24/*16	16/*11	24/*23	21/*13	14/*9	24/*20	18/*12	12/*8	24/*17	16/*10	11/*7	24/*15	14/*9	9/*6	21/*14	12/*8	8/*5
		6	26.6	Non-Exposed / *Exposed	26	26	26/*22	26	26	26/*19	26	26/*25	26/*17	26	26/*22	26/*15	26	26/*20	23/*14	26	26/*19	20/*13	26	24/*16	16/*11	26/*23	21/*13	14/*9	26/*20	18/*12	12/*8	26/*17	16/*10	10/*7	24/*15	14/*9	9/*6	21/*13	12/*8	8/*5
		8	33.7	Non-Exposed / *Exposed	29	29	29/*28	29	29	29/*25	29	29/*22	29	29/*28	29/*19	29	29/*26	29/*17	29/*26	29/*23	26/*15	29/*22	29/*20	21/*13	29/*19	26/*17	17/*10	25/*16	22/*15	15/*9	22/*14	20/*13	12/*8	19/*12	17/*11	11/*7	17/*11	15/*10	9/*6	
		10	39.8	Non-Exposed / *Exposed	35	35	35/*28	35	35/*34	35/*24	35/*34	35/*31	35/*29	35/*31	35/*28	32/*19	35/*28	35/*25	35/*23	36/*15	34/*22	30/*20	21/*13	29/*19	26/*17	17/*10	25/*16	22/*14	15/*9	22/*14	19/*13	12/*8	19/*12	17/*11	11/*7	17/*11	15/*10	9/*6		
		12	45.0	Non-Exposed / *Exposed	42	42/*37	42/*27	42/*38	42/*34	40/*24	42/*34	42/*30	36/*21	42/*31	42/*27	32/*19	42/*28	38/*25	28/*17	39/*26	35/*23	25/*15	33/*22	30/*19	21/*12	28/*19	25/*17	17/*10	24/*16	22/*14	14/*9	21/*14	19/*13							

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

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Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

Roof Tech - Mini Gen II - ASCE 7-16 Hip - Flush-Mount - 15 Foot Max Mean Roof Height - OSB Only - Portrait 72 Cell - 60 mm Long Screws

Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->			Roof Area ---->			95			100			105			110			115			120			130			140			150			160			170			180		
										1			1			1			1			1			1			1			1			1			1			1			1		
										2			2			2			2			2			2			2			2			2			2			2			2		
25	B	2	9.5	Non-Exposed / *Exposed	31	31/*24	31/*22	31/*30	31/*21	31/*20	31/*27	31/*19	28/*18	31/*24	28/*17	25/*16	31/*22	25/*16	23/*14	31/*20	23/*14	21/*13	26/*16	19/*12	17/*11	22/*14	16/*10	15/*9	19/*12	14/*9	12/*8	16/*10	12/*8	11/*7	14/*9	10/*7	9/*6	13/*8	9/*6	8/*5					
		3	14.0	Non-Exposed / *Exposed	30	30/*24	30/*22	30	30/*21	30/*19	30/*26	30/*19	28/*17	30/*24	27/*17	25/*16	30/*22	25/*16	22/*14	30/*20	22/*14	20/*13	26/*16	19/*12	17/*11	22/*14	16/*10	14/*9	19/*12	14/*9	12/*8	16/*10	12/*8	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5					
		4	18.4	Non-Exposed / *Exposed	30	30/*24	30/*22	30/*29	30/*21	30/*19	30/*26	30/*19	27/*17	30/*24	27/*17	25/*16	30/*21	24/*16	22/*14	30/*19	22/*14	20/*13	26/*16	19/*12	17/*11	22/*14	16/*10	14/*9	19/*12	13/*9	12/*8	16/*10	12/*7	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5					
		5	22.6	Non-Exposed / *Exposed	31	31/*29	NA	31	31/*26	NA	31	31/*23	NA	31	31/*21	NA	31/*28	30/*19	NA	31/*26	27/*17	NA	31/*21	23/*14	NA	29/*18	19/*12	NA	24/*16	16/*11	NA	21/*13	14/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA					
		6	26.6	Non-Exposed / *Exposed	33	33/*29	NA	33	33/*26	NA	33	33/*23	NA	33/*31	33/*21	NA	33/*28	29/*19	NA	33/*25	27/*17	NA	33/*21	22/*14	NA	28/*18	19/*12	NA	24/*15	16/*10	NA	21/*13	14/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA					
		8	33.7	Non-Exposed / *Exposed	36	32/*19	26/*15	36/*35	28/*17	22/*13	36/*31	24/*16	19/*12	36/*28	22/*14	17/*11	36/*25	20/*13	15/*10	36/*23	18/*12	14/*9	30/*19	15/*10	12/*7	26/*16	13/*8	10/*6	9/*85	19/*12	10/*6	7/*5	17/*11	9/*5	6/*4	15/*10	8/*5	6/*4							
		10	39.8	Non-Exposed / *Exposed	40/*39	31/*19	26/*15	40/*34	27/*17	21/*13	40/*30	24/*15	18/*12	40/*27	22/*14	17/*11	40/*25	20/*13	15/*10	35/*23	18/*12	14/*9	30/*19	15/*10	12/*7	25/*16	13/*8	10/*6	22/*14	11/*7	8/*5	19/*12	10/*6	7/*5	17/*11	8/*5	6/*4	15/*9	7/*5	6/*4					
		12	45.0	Non-Exposed / *Exposed	45/*37	31/*19	25/*15	45/*33	26/*17	21/*13	45/*30	24/*15	18/*12	45/*27	21/*14	16/*11	39/*25	19/*13	15/*10	35/*22	18/*12	14/*9	29/*19	15/*10	11/*7	25/*16	13/*8	10/*6	21/*14	11/*7	8/*5	19/*12	10/*6	7/*5	16/*11	8/*5	6/*4	15/*9	7/*5	6/*4					
	C	2	9.5	Non-Exposed / *Exposed	31/*27	31/*19	28/*18	31/*24	27/*17	25/*16	31/*21	24/*15	18/*12	24/*15	22/*14	31/*19	22/*14	20/*13	28/*17	20/*13	18/*12	25/*16	18/*11	17/*11	21/*13	15/*10	14/*9	18/*11	13/*8	12/*7	15/*10	11/*7	10/*6	13/*8	10/*6	9/*6	11/*7	8/*5	8/*5	10/*6	7/*5	7/*4			
		3	14.0	Non-Exposed / *Exposed	30/*27	30/*19	28/*18	30/*24	27/*17	25/*16	30/*21	24/*15	18/*12	22/*14	30/*19	22/*14	20/*13	27/*17	20/*13	18/*12	25/*16	18/*11	16/*10	21/*13	15/*10	14/*9	18/*11	13/*8	12/*7	15/*10	11/*7	10/*6	13/*8	9/*6	9/*6	11/*7	8/*5	8/*5	10/*6	7/*5	7/*4				
		4	18.4	Non-Exposed / *Exposed	30/*26	30/*19	28/*17	30/*24	27/*17	25/*16	30/*21	24/*15	18/*12	22/*14	30/*19	22/*14	20/*13	27/*17	20/*13	18/*11	25/*16	18/*11	16/*10	21/*13	15/*10	14/*9	17/*11	13/*8	12/*7	15/*10	11/*7	10/*6	13/*8	9/*6	9/*6	11/*7	8/*5	8/*5	10/*6	7/*5	7/*4				
		5	22.6	Non-Exposed / *Exposed	31	31/*23	NA	31	31/*21	NA	31/*28	29/*19	NA	31	31/*25	26/*17	NA	31/*23	24/*15	NA	31/*21	22/*14	NA	27/*17	18/*12	NA	23/*15	15/*10	NA	20/*13	13/*9	NA	17/*11	12/*7	NA	15/*10	10/*6	NA	13/*8	9/*6	NA				
		6	26.6	Non-Exposed / *Exposed	33	33/*23	NA	33/*31	32/*21	NA	33/*27	29/*18	NA	33/*25	26/*17	NA	33/*22	24/*15	NA	32/*20	22/*14	NA	27/*17	18/*12	NA	23/*15	15/*10	NA	19/*13	13/*8	NA	17/*11	11/*7	NA	15/*10	10/*6	NA	13/*8	9/*6	NA					
		8	33.7	Non-Exposed / *Exposed	36/*31	24/*16	19/*12	36/*28	22/*14	17/*11	36/*25	20/*13	15/*10	35/*23	18/*11	14/*9	32/*20	16/*10	12/*8	29/*19	15/*9	11/*7	24/*16	12/*8	9/*6	21/*13	11/*7	8/*5	12/*8	9/*6	7/*4	16/*10	8/*5	6/*4	14/*9	7/*4	5/*3	12/*8	6/*4	5/*3					
		10	39.8	Non-Exposed / *Exposed	40/*31	24/*16	19/*12	40/*27	22/*14	16/*11	39/*25	19/*13	15/*10	35/*22	18/*11	13/*9	31/*20	16/*10	12/*8	29/*18	15/*9	11/*7	24/*16	12/*8	9/*6	20/*13	10/*7	8/*5	18/*11	9/*6	7/*4	15/*10	8/*5	6/*4	13/*9	7/*4	5/*3	12/*8	6/*4	5/*3					
		12	45.0	Non-Exposed / *Exposed	45/*30	24/*15	18/*12	45/*27	21/*14	16/*11	38/*24	19/*12	15/*10	34/*22	17/*11	13/*9	31/*20	16/*10	12/*8	28/*18	14/*9	11/*7	24/*15	12/*8	9/*6	20/*13	10/*7	8/*5	17/*11	9/*6	7/*4	15/*10	8/*5	6/*4	13/*9	7/*4	5/*3	12/*8	6/*4	5/*3					
	D	2	9.5	Non-Exposed / *Exposed	31/*22	25/*16	23/*14	31/*19	22/*14	20/*13	27/*17	20/*12	18/*11	25/*16	18/*11	16/*10	22/*14	16/*10	15/*9	20/*13	15/*9	13/*9	17/*11	12/*8	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5	11/*7	8/*5	7/*4	9/*6	7/*4	6/*4	8/*5	6/*4	5/*3					
		3	14.0	Non-Exposed / *Exposed	30/*21	24/*16	22/*14	30/*19	22/*14	20/*13	27/*17	20/*12	18/*11	24/*15	18/*11	16/*10	22/*14	16/*10	15/*9	20/*13	14/*9	13/*8	17/*11	12/*8	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5	11/*7	8/*5	7/*4	9/*6	7/*4	6/*4	8/*5	6/*4	5/*3					
		4	18.4	Non-Exposed / *Exposed	30/*21	24/*15	22/*14	30/*19	22/*14	20/*13	27/*17	19/*12	18/*11	24/*15	17/*11	16/*10	22/*14	16/*10	14/*9	20/*13	14/*9	13/*8	17/*11	12/*8	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5	11/*7	8/*5	7/*4	9/*6	7/*4	6/*4	8/*5	6/*4	5/*3					
		5	22.6	Non-Exposed / *Exposed	31/*28	30/*19	NA	31/*25	26/*17	NA	31/*22	24/*15	NA	31/*20	21/*14	NA	29/*18	19/*12	NA	26/*17	18/*11	NA	22/*14	15/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA	14/*9	9/*6	NA	12/*8	8/*5	NA	11/*7	7/*5	NA					
		6	26.6	Non-Exposed / *Exposed	33/*28	29/*19	NA	33/*25	26/*17	NA	33/*22	23/*15	NA	31/*20	21/*14	NA	28/*18	19/*12	NA	26/*17	17/*11	NA	22/*14	15/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA	14/*9	9/*6	NA	12/*8	8/*5	NA	11/*7	7/*5	NA					
		8	33.7	Non-Exposed / *Exposed	36/*25	20/*13	15/*10	35/*23	18/*11	14/*9	32/*20	16/*10	12/*8	29/*18	14/*9	11/*7	26/*17	13/*8	10/*6	24/*15	12/*8	9/*6	20/*13	10/*6	8/*5	17/*11	9/*5	6/*4	14/*9	7/*5	6/*4	13/*8	6/*4	5/*3	11/*7	6/*4	4/*3	10/*6	5/*3	4/*2					
		10	39.8	Non-Exposed / *Exposed	40/*25	20/*13	15/*10	34/*22	18/*11	13/*9	31/*20	16/*10	12/*8	28/*18	14/*9	11/*7	25/*16	13/*8	10/*6	23/*15	12/*8	9/*6	20/*13	10/*6	8/*5	17/*11	9/*5	6/*4	14/*9	7/*5	6/*4	12/*8	6/*4	5/*3	11/*7	6/*4	4/*3	10/*6	5/*3	4/*2					
		12	45.0	Non-Exposed / *Exposed	39/*24	19/*13	15/*10	34/*22	17/*11	13/*9	30/*20	16/*10	12/*8	28/*18	14/*9	11/*7	25/*16	13/*8	10/*6	23/*15	12/*8	9/*6	19/*13	10/*6	8/*5	16/*11	8/*5	6/*4	14/*9	7/*5	6/*4	12/*8	6/*4	5/*3	11/*7	6/*4	4/*3	10/*6	5/*3	4/*2					
30	B	2	9.5	Non-Exposed / *Exposed	28	28/*24	28/*22	28	28/*21	28/*20	28/*27	28/*19	28/*18	28/*24	28/*17	25/*16	28/*22	25/*16	23/*14	28/*20	23/*14	21/*13	26/*16	19/*12	17/*11	22/*14	16/*10	15/*9	19/*12	14/*9	12/*8	16/*10	12/*8	11/*7	14/*9	10/*7	9/*6	13/*8	9/*6	8/*5					
		3	14.0	Non-Exposed / *Exposed	26	26/*24	26/*22	26	26/*21	26/*19	26	26/*19	26/*17	26/*24	26/*17	25/*16	26/*22	25/*16	22/*14	26/*20	22/*14	20/*13	26/*16	19/*12	17/*11	22/*14	16/*10	14/*9	19/*12	14/*9	12/*8	16/*10	12/*8	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5					
		4	18.4	Non-Exposed / *Exposed	27	27/*24	27/*22	27	27/*21	27/*19	27/*26	27/*19	27/*17	27/*24	27/*17	25/*16	27/*21	24/*16	22/*14	27/*19	22/*14	20/*13	26/*16	19/*12	17/*11	22/*14	16/*10	14/*9	19/*12	13/*9	12/*8	16/*10	12/*7	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5					
		5	22.6	Non-Exposed / *Exposed	28	28	NA	28	28/*26	NA	28	28/*23	NA	28	28/*21	NA	28	28/*19	NA	28/*26	27/*17	NA	28/*21	23/*14	NA	28/*18	19/*12	NA	24/*16	16/*11	NA	21/*13	14/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA					
		6	26.6	Non-Exposed / *Exposed	29	29	NA	29	29/*26	NA	29	29/*23	NA	29	29/*21																														

Roof Tech - Mini Gen II - ASCE 7-16 Hip - Flush-Mount - 15 Foot Max Mean Roof Height - OSB Only - Portrait 72 Cell - 60 mm Long Screws																																								
Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->	95			100			105			110			115			120			130			140			150			160			170			180		
				Roof Area ---->	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
40	B	2	9.5	Non-Exposed / *Exposed	21	21	21	21	21	21/*20	21	21/*19	21/*18	21	21/*17	21/*16	21	21/*16	21/*14	21/*20	21/*14	21/*13	21/*16	19/*12	17/*11	21/*14	16/*10	15/*9	19/*12	14/*9	12/*8	16/*10	12/*8	11/*7	14/*9	10/*7	9/*6	13/*8	9/*6	8/*5
		3	14.0	Non-Exposed / *Exposed	20	20	20	20	20	20/*19	20	20/*19	20/*17	20	20/*17	20/*16	20	20/*16	20/*14	20	20/*14	20/*13	20/*16	19/*12	17/*11	20/*14	16/*10	14/*9	19/*12	14/*9	12/*8	16/*10	12/*8	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5
		4	18.4	Non-Exposed / *Exposed	21	21	21	21	21	21/*19	21	21/*19	21/*17	21	21/*17	21/*16	21	21/*16	21/*14	21/*19	21/*14	20/*13	21/*16	19/*12	17/*11	21/*14	16/*10	14/*9	19/*12	13/*9	12/*8	16/*10	12/*7	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5
		5	22.6	Non-Exposed / *Exposed	22	22	NA	22	22	NA	22	22	NA	22	22/*21	NA	22	22/*19	NA	22	22/*17	NA	22/*21	22/*14	NA	22/*18	19/*12	NA	22/*16	16/*11	NA	21/*13	14/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA
		6	26.6	Non-Exposed / *Exposed	24	24	NA	24	24	NA	24	24/*23	NA	24	24/*21	NA	24	24/*19	NA	24	24/*17	NA	24/*21	22/*14	NA	24/*18	19/*12	NA	24/*15	16/*10	NA	21/*13	14/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA
		8	33.7	Non-Exposed / *Exposed	28	28/*19	26/*15	28	28/*17	22/*13	28	24/*16	19/*12	28	22/*14	17/*11	28/*25	20/*13	15/*10	28/*23	18/*12	14/*9	28/*19	15/*10	12/*7	26/*16	13/*8	10/*6	9/*85	19/*12	10/*6	7/*5	17/*11	9/*85	6/*4	15/*10	8/*5	6/*4		
		10	39.8	Non-Exposed / *Exposed	32	31/*19	26/*15	32	27/*17	21/*13	32/*30	24/*15	18/*12	32/*27	22/*14	17/*11	32/*25	20/*13	15/*10	32/*23	18/*12	14/*9	30/*19	15/*10	12/*7	25/*16	13/*8	10/*6	22/*14	11/*7	8/*85	19/*12	10/*6	7/*5	17/*11	8/*85	6/*4	15/*9	7/*5	6/*4
		12	45.0	Non-Exposed / *Exposed	36	31/*19	25/*15	36/*33	26/*17	21/*13	36/*30	24/*15	18/*12	36/*27	21/*14	16/*11	36/*25	19/*13	15/*10	35/*22	18/*12	14/*9	29/*19	15/*10	11/*7	25/*16	13/*8	10/*6	21/*14	11/*7	8/*85	19/*12	10/*6	7/*5	16/*11	8/*85	6/*4	15/*9	7/*5	6/*4
	C	2	9.5	Non-Exposed / *Exposed	21	21/*19	21/*18	21	21/*17	21/*16	21	21/*15	21/*14	21/*19	21/*14	20/*13	21/*17	20/*13	18/*12	21/*16	18/*11	17/*11	21/*13	15/*10	14/*9	18/*11	13/*8	12/*7	15/*10	11/*7	10/*6	13/*8	10/*6	9/*6	11/*7	8/*85	8/*85	10/*6	7/*5	7/*4
		3	14.0	Non-Exposed / *Exposed	20	20/*19	20/*18	20	20/*17	20/*16	20	20/*15	20/*14	20/*19	20/*14	20/*13	20/*17	20/*13	18/*12	20/*16	18/*11	16/*10	20/*13	15/*10	14/*9	18/*11	13/*8	12/*7	15/*10	11/*7	10/*6	13/*8	9/*6	9/*6	11/*7	8/*85	8/*85	10/*6	7/*5	7/*4
		4	18.4	Non-Exposed / *Exposed	21	21/*19	21/*17	21	21/*17	21/*16	21	21/*15	21/*14	21/*19	21/*14	20/*13	21/*17	20/*13	18/*11	21/*16	18/*11	16/*10	21/*13	15/*10	14/*9	17/*11	13/*8	12/*7	15/*10	11/*7	10/*6	13/*8	9/*6	9/*6	11/*7	8/*85	8/*85	10/*6	7/*5	7/*4
		5	22.6	Non-Exposed / *Exposed	22	22	NA	22	22/*21	NA	22	22/*19	NA	22	22/*17	NA	22	22/*15	NA	22/*21	22/*14	NA	22/*17	18/*12	NA	22/*15	15/*10	NA	20/*13	13/*9	NA	17/*11	12/*7	NA	15/*10	10/*6	NA	13/*8	9/*6	NA
		6	26.6	Non-Exposed / *Exposed	24	24/*23	NA	24	24/*21	NA	24	24/*18	NA	24	24/*17	NA	24/*22	24/*15	NA	24/*20	22/*14	NA	24/*17	18/*12	NA	23/*15	15/*10	NA	19/*13	13/*8	NA	17/*11	11/*7	NA	15/*10	10/*6	NA	13/*8	9/*6	NA
		8	33.7	Non-Exposed / *Exposed	28	24/*16	19/*12	28	22/*14	17/*11	28/*25	20/*13	15/*10	28/*23	18/*11	14/*9	28/*20	16/*10	12/*8	28/*19	15/*9	11/*7	24/*16	12/*8	9/*6	21/*13	11/*7	8/*85	12/*8	9/*6	7/*4	16/*10	8/*85	6/*4	14/*9	7/*4	5/*3	12/*8	6/*4	5/*3
		10	39.8	Non-Exposed / *Exposed	32/*31	24/*16	19/*12	32/*27	22/*14	16/*11	32/*25	19/*13	15/*10	32/*22	18/*11	13/*9	31/*20	16/*10	12/*8	29/*18	15/*9	11/*7	24/*16	12/*8	9/*6	20/*13	10/*7	8/*85	18/*11	9/*6	7/*4	15/*10	8/*85	6/*4	13/*9	7/*4	5/*3	12/*8	6/*4	5/*3
		12	45.0	Non-Exposed / *Exposed	36/*30	24/*15	18/*12	36/*27	21/*14	16/*11	36/*24	19/*12	15/*10	34/*22	17/*11	13/*9	31/*20	16/*10	12/*8	28/*18	14/*9	11/*7	24/*15	12/*8	9/*6	20/*13	10/*7	8/*85	17/*11	9/*6	7/*4	15/*10	8/*85	6/*4	13/*9	7/*4	5/*3	12/*8	6/*4	5/*3
	D	2	9.5	Non-Exposed / *Exposed	21	21/*16	21/*14	21/*19	21/*14	20/*13	21/*17	20/*12	18/*11	21/*16	18/*11	16/*10	21/*14	16/*10	15/*9	20/*13	15/*9	13/*9	17/*11	12/*8	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*85	11/*7	8/*85	7/*4	9/*6	7/*4	6/*4	8/*85	6/*4	5/*3
		3	14.0	Non-Exposed / *Exposed	20	20/*16	20/*14	20/*19	20/*14	20/*13	20/*17	20/*12	18/*11	20/*15	18/*11	16/*10	20/*14	16/*10	15/*9	20/*13	14/*9	13/*8	17/*11	12/*8	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*85	11/*7	8/*85	7/*4	9/*6	7/*4	6/*4	8/*85	6/*4	5/*3
		4	18.4	Non-Exposed / *Exposed	21	21/*15	21/*14	21/*19	21/*14	20/*13	21/*17	19/*12	18/*11	21/*16	16/*10	21/*14	16/*10	14/*9	20/*13	14/*9	13/*8	17/*11	12/*8	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*85	11/*7	8/*85	7/*4	9/*6	7/*4	6/*4	8/*85	6/*4	5/*3	
		5	22.6	Non-Exposed / *Exposed	22	22/*19	NA	22	22/*17	NA	22	22/*15	NA	22/*20	21/*14	NA	22/*18	19/*12	NA	22/*17	18/*11	NA	22/*14	15/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA	14/*9	9/*6	NA	12/*8	8/*85	NA	11/*7	7/*5	NA
		6	26.6	Non-Exposed / *Exposed	24	24/*19	NA	24	24/*17	NA	24/*22	23/*15	NA	24/*20	21/*14	NA	24/*18	19/*12	NA	24/*17	17/*11	NA	22/*14	15/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA	14/*9	9/*6	NA	12/*8	8/*85	NA	11/*7	7/*5	NA
		8	33.7	Non-Exposed / *Exposed	28/*25	20/*13	15/*10	28/*23	18/*11	14/*9	28/*20	16/*10	12/*8	28/*18	14/*9	11/*7	26/*17	13/*8	10/*6	24/*15	12/*8	9/*6	20/*13	10/*6	8/*85	17/*11	9/*5	6/*4	14/*9	7/*5	6/*4	13/*8	6/*4	5/*3	11/*7	6/*4	4/*3	10/*6	5/*3	4/*2
		10	39.8	Non-Exposed / *Exposed	32/*25	20/*13	15/*10	32/*22	18/*11	13/*9	31/*20	16/*10	12/*8	28/*18	14/*9	11/*7	25/*16	13/*8	10/*6	23/*15	12/*8	9/*6	20/*13	10/*6	8/*85	17/*11	9/*5	6/*4	14/*9	7/*5	6/*4	12/*8	6/*4	5/*3	11/*7	6/*4	4/*3	10/*6	5/*3	4/*2
		12	45.0	Non-Exposed / *Exposed	36/*24	19/*13	15/*10	34/*22	17/*11	13/*9	30/*20	16/*10	12/*8	28/*18	14/*9	11/*7	25/*16	13/*8	10/*6	23/*15	12/*8	9/*6	19/*13	10/*6	8/*85	16/*11	8/*85	6/*4	14/*9	7/*5	6/*4	12/*8	6/*4	5/*3	11/*7	6/*4	4/*3	10/*6	5/*3	4/*2
50	B	2	9.5	Non-Exposed / *Exposed	17	17	17	17	17	17	17	17	17	17	17	17/*16	17	17/*16	17/*14	17	17/*14	17/*13	17/*16	17/*12	17/*11	17/*14	16/*10	15/*9	17/*12	14/*9	12/*8	16/*10	12/*8	11/*7	14/*9	10/*7	9/*6	13/*8	9/*6	8/*5
		3	14.0	Non-Exposed / *Exposed	17	17	17	17	17	17	17	17	17	17	17	17/*16	17	17/*16	17/*14	17	17/*14	17/*13	17/*16	17/*12	17/*11	17/*14	16/*10	14/*9	17/*12	14/*9	12/*8	16/*10	12/*8	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5
		4	18.4	Non-Exposed / *Exposed	17	17	17	17	17	17	17	17	17	17	17	17/*16	17	17/*16	17/*14	17	17/*14	17/*13	17/*16	17/*12	17/*11	17/*14	16/*10	14/*9	17/*12	13/*9	12/*8	16/*10	12/*7	11/*7	14/*9	10/*7	9/*6	12/*8	9/*6	8/*5
		5	22.6	Non-Exposed / *Exposed	18	18	NA	18	18	NA	18	18	NA	18	18	NA	18	18/*17	NA	18	18/*14	NA	18	18/*12	NA	18	18/*12	NA	18/*16	16/*11	NA	18/*13	14/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA
		6	26.6	Non-Exposed / *Exposed	19	19	NA	19	19	NA	19	19	NA	19	19	NA	19	19/*17	NA	19	19/*14	NA	19	19/*12	NA	19	19/*12	NA	19/*15	16/*10	NA	18/*13	14/*9	NA	18/*12	12/*8	NA	16/*10	11/*7	NA
		8	33.7	Non-Exposed / *Exposed	23	23/*19	23/*15	23	23/*17	22/*13	23	23/*16	19/*12	23	22/*14	17/*11	23	20/*13	15/*10	23	18/*12	14/*9	23/*19	15/*10	12/*7	23/*16	13/*8	10/*6	22/*14	11/*7	9/*85	19/*12	10/*6	7/*5	17/*11	9/*85	6/*4	15/*10	8/*85	6/*4
		10	39.8	Non-Exposed / *Exposed	28	28/*19	26/*15	28	27/*17	21/*13	28	24/*15	18/*12	28/*27	22/*14	17/*11	28/*25	20/*13	15/*10	28/*23	18/*12	14/*9	28/*19	15/*10																

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

Roof Tech - Mini Gen II - ASCE 7-16 Hip - Flush-Mount - 15 Foot Max Mean Roof Height - Plywood Only - Portrait 72 Cell - 60 mm Long Screws																																								
Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->	95			100			105			110			115			120			130			140			150			160			170			180		
				Roof Area ---->	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
0	B	2	9.5	Non-Exposed / *Exposed	68 /*42	55 /*29	47 /*27	68 /*36	45 /*26	38 /*24	58 /*32	37 /*23	34 /*21	50 /*29	33 /*21	30 /*19	43 /*26	30 /*19	28 /*17	38 /*24	27 /*17	25 /*16	32 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7
		3	14.0	Non-Exposed / *Exposed	67 /*41	54 /*29	46 /*27	66 /*36	44 /*26	38 /*24	56 /*32	37 /*23	34 /*21	49 /*29	33 /*21	30 /*19	43 /*26	30 /*19	27 /*17	38 /*24	27 /*17	25 /*16	31 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7
		4	18.4	Non-Exposed / *Exposed	65 /*41	52 /*29	45 /*26	64 /*36	43 /*26	37 /*24	55 /*32	36 /*23	33 /*21	48 /*29	33 /*21	30 /*19	42 /*26	30 /*19	27 /*17	37 /*24	27 /*17	25 /*16	31 /*20	23 /*14	21 /*13	26 /*17	19 /*12	18 /*11	23 /*14	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	11 /*7	15 /*10	11 /*7	10 /*7
		5	22.6	Non-Exposed / *Exposed	64	64 /*35	NA	64 /*56	61 /*31	NA	64 /*45	49 /*28	NA	64 /*38	41 /*25	NA	64 /*34	36 /*23	NA	61 /*31	33 /*21	NA	43 /*26	27 /*18	NA	35 /*22	23 /*15	NA	30 /*19	20 /*13	NA	26 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA
		6	26.6	Non-Exposed / *Exposed	63	63 /*35	NA	63 /*54	59 /*31	NA	63 /*44	48 /*28	NA	63 /*37	40 /*25	NA	63 /*34	36 /*23	NA	59 /*31	32 /*21	NA	42 /*26	27 /*17	NA	34 /*22	23 /*15	NA	29 /*19	20 /*13	NA	25 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA
		8	33.7	Non-Exposed / *Exposed	61 /*54	43 /*24	35 /*18	61 /*44	35 /*21	29 /*16	61 /*37	31 /*19	25 /*15	61 /*34	27 /*17	21 /*13	57 /*31	24 /*16	18 /*12	48 /*28	22 /*14	17 /*11	36 /*23	19 /*12	14 /*9	31 /*20	16 /*10	12 /*8	27 /*17	14 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*12	9 /*6	7 /*4
	C	10	39.8	Non-Exposed / *Exposed	59 /*52	41 /*23	34 /*18	59 /*43	35 /*21	28 /*16	59 /*37	30 /*19	24 /*14	59 /*33	26 /*17	21 /*13	54 /*30	24 /*15	18 /*12	46 /*27	22 /*14	17 /*11	36 /*23	18 /*12	14 /*9	30 /*20	16 /*10	12 /*8	26 /*17	14 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*12	9 /*6	7 /*4
		12	45.0	Non-Exposed / *Exposed	57 /*49	40 /*23	33 /*18	57 /*41	34 /*21	28 /*16	57 /*36	29 /*19	24 /*14	57 /*33	26 /*17	20 /*13	52 /*30	24 /*15	18 /*12	44 /*27	22 /*14	17 /*11	35 /*23	18 /*12	14 /*9	30 /*20	16 /*10	12 /*8	26 /*17	13 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*12	9 /*6	7 /*4
		2	9.5	Non-Exposed / *Exposed	58 /*33	38 /*24	34 /*22	50 /*29	33 /*21	30 /*19	43 /*26	30 /*19	27 /*17	37 /*23	27 /*17	24 /*16	34 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	25 /*16	18 /*12	17 /*11	21 /*14	16 /*10	14 /*9	18 /*12	13 /*9	12 /*8	16 /*10	12 /*7	11 /*7	14 /*9	10 /*7	9 /*6	12 /*8	9 /*6	8 /*5
		3	14.0	Non-Exposed / *Exposed	57 /*32	37 /*23	34 /*21	49 /*29	33 /*21	30 /*19	42 /*26	29 /*19	27 /*17	37 /*23	27 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	25 /*16	18 /*12	17 /*11	21 /*14	16 /*10	14 /*9	18 /*12	13 /*9	12 /*8	16 /*10	12 /*7	11 /*7	14 /*9	10 /*7	9 /*6	12 /*8	9 /*6	8 /*5
		4	18.4	Non-Exposed / *Exposed	56 /*32	37 /*23	33 /*21	48 /*29	33 /*21	30 /*19	41 /*26	29 /*19	27 /*17	36 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	25 /*16	18 /*12	17 /*11	21 /*14	15 /*10	14 /*9	18 /*12	13 /*9	12 /*8	16 /*10	12 /*7	11 /*7	14 /*9	10 /*7	9 /*6	12 /*8	9 /*6	8 /*5
		5	22.6	Non-Exposed / *Exposed	64 /*46	50 /*28	NA	64 /*38	41 /*25	NA	64 /*34	35 /*23	NA	58 /*30	32 /*20	NA	48 /*27	29 /*19	NA	41 /*25	26 /*17	NA	33 /*21	22 /*14	NA	28 /*18	19 /*12	NA	24 /*15	16 /*10	NA	21 /*13	14 /*9	NA	18 /*12	12 /*8	NA	16 /*10	11 /*7	NA
10	B	6	26.6	Non-Exposed / *Exposed	63 /*45	49 /*28	NA	63 /*37	40 /*25	NA	63 /*33	35 /*22	NA	56 /*30	32 /*20	NA	47 /*27	29 /*18	NA	40 /*25	26 /*17	NA	32 /*21	22 /*14	NA	28 /*18	19 /*12	NA	24 /*15	16 /*10	NA	21 /*13	14 /*9	NA	18 /*12	12 /*8	NA	16 /*10	11 /*7	NA
		8	33.7	Non-Exposed / *Exposed	61 /*38	31 /*19	25 /*15	61 /*34	26 /*17	21 /*13	55 /*30	24 /*15	18 /*12	46 /*27	22 /*14	16 /*11	39 /*25	20 /*13	15 /*10	35 /*23	18 /*12	14 /*9	30 /*19	15 /*10	12 /*7	25 /*16	13 /*8	10 /*6	22 /*14	11 /*7	8 /*5	19 /*12	10 /*6	7 /*5	17 /*11	8 /*5	6 /*4	15 /*9	7 /*5	6 /*4
		10	39.8	Non-Exposed / *Exposed	59 /*37	30 /*19	24 /*14	59 /*33	26 /*17	20 /*13	53 /*30	24 /*15	18 /*12	44 /*27	21 /*14	16 /*11	38 /*24	19 /*13	15 /*10	34 /*22	18 /*12	14 /*9	29 /*19	15 /*10	11 /*7	25 /*16	13 /*8	10 /*6	21 /*14	11 /*7	8 /*5	19 /*12	10 /*6	7 /*5	16 /*11	8 /*5	6 /*4	15 /*9	7 /*5	6 /*4
		12	45.0	Non-Exposed / *Exposed	57 /*36	30 /*19	24 /*14	57 /*32	26 /*17	20 /*13	50 /*29	23 /*15	18 /*12	43 /*27	21 /*14	16 /*11	37 /*24	19 /*13	15 /*10	34 /*22	18 /*11	13 /*9	29 /*19	15 /*10	11 /*7	24 /*16	13 /*8	10 /*6	21 /*14	11 /*7	8 /*5	18 /*12	10 /*6	7 /*5	16 /*11	8 /*5	6 /*4	14 /*9	7 /*5	6 /*4
	C	2	9.5	Non-Exposed / *Exposed	43 /*26	30 /*19	27 /*17	37 /*23	27 /*17	24 /*16	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	27 /*17	20 /*12	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4
		3	14.0	Non-Exposed / *Exposed	42 /*26	30 /*19	27 /*17	37 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	21 /*14	20 /*13	27 /*17	19 /*12	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4
		4	18.4	Non-Exposed / *Exposed	42 /*26	29 /*19	27 /*17	36 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	29 /*19	21 /*14	19 /*13	27 /*17	19 /*12	18 /*11	24 /*15	18 /*11	16 /*10	20 /*13	15 /*9	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4
		5	22.6	Non-Exposed / *Exposed	64 /*34	36 /*23	NA	58 /*30	32 /*20	NA	47 /*27	29 /*18	NA	39 /*24	26 /*17	NA	35 /*22	23 /*15	NA	32 /*20	21 /*14	NA	26 /*17	18 /*12	NA	22 /*14	15 /*10	NA	19 /*12	13 /*8	NA	17 /*11	11 /*7	NA	15 /*9	10 /*6	NA	13 /*8	9 /*6	NA
		6	26.6	Non-Exposed / *Exposed	63 /*34	35 /*23	NA	56 /*30	32 /*20	NA	46 /*27	28 /*18	NA	38 /*24	26 /*16	NA	34 /*22	23 /*15	NA	31 /*20	21 /*14	NA	26 /*17	18 /*11	NA	22 /*14	15 /*10	NA	19 /*12	13 /*8	NA	17 /*11	11 /*7	NA	15 /*9	10 /*6	NA	13 /*8	9 /*6	NA
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20	B	10	39.8	Non-Exposed / *Exposed	53 /*30	24 /*15	18 /*12	44 /*27	21 /*14	16 /*11	37 /*24	19 /*12	15 /*10	34 /*22	17 /*11	13 /*9	31 /*20	16 /*10	12 /*8	28 /*18	14 /*9	11 /*7	24 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	17 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3
		12	45.0	Non-Exposed / *Exposed	51 /*30	23 /*15	18 /*12	43 /*26	21 /*14	16 /*11	37 /*24	19 /*12	15 /*9	33 /*22	17 /*11	13 /*9	30 /*20	16 /*10	12 /*8	28 /*18	14 /*9	11 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	17 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3
	C	2	9.5	Non-Exposed / *Exposed	68 /*42	55 /*29	47 /*27	68 /*36	45 /*26	38 /*24	58 /*32	37 /*23	34 /*21	50 /*29	33 /*21	30 /*19	43 /*26	30 /*19	28 /*17	38 /*24	27 /*17	25 /*16	32 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7
		3	14.0	Non-Exposed / *Exposed	67 /*41	54 /*29	46 /*27	66 /*36	44 /*26	38 /*24	56 /*32	37 /*23	34 /*21	49 /*29	33 /*21	30 /*19	43 /*26	30 /*19	27 /*17	38 /*24	27 /*17	25 /*16	31 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7
		4	18.4	Non-Exposed / *Exposed	65 /*41	52 /*29	45 /*26	64 /*36	43 /*26	37 /*24	55 /*32	36 /*23	33 /*21	48 /*29	33 /*21	30 /*19	42 /*26	30 /*19	27 /*17	37 /*24	27 /*17	25 /*16	31 /*20	23 /*14	21 /*13	26 /*17	19 /*12	18 /*11	23 /*14	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	11 /*7	15 /*10	11 /*7	10 /*7
		5	22.6	Non-Exposed / *Exposed	64	64 /*35	NA	64 /*56	61 /*31	NA	64 /*45	49 /*28	NA	64 /*38	41 /*25	NA	64 /*34	36 /*23	NA	61 /*31	33 /*21	NA	43 /*26	27 /*18	NA	35 /*22	23 /*15	NA	30 /*19	20 /*13	NA	26 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA
		6	26.6	Non-Exposed / *Exposed	63	63 /*35	NA	63 /*54	59 /*31	NA	63 /*44	48 /*28	NA	63 /*37	40 /*25	NA	63 /*34	36 /*23	NA	59 /*31	32 /*21	NA	42 /*26	27 /*17	NA	34 /*22	23 /*15	NA	29 /*19	20 /*13	NA	25 /*16	17 /*11	NA	22 /*14	15 /*10	NA			

Roof Tech - Mini Gen II - ASCE 7-16 Hip - Flush-Mount - 15 Foot Max Mean Roof Height - Plywood Only - Portrait 72 Cell - 60 mm Long Screws																																								
Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->	95			100			105			110			115			120			130			140			150			160			170			180		
				Roof Area ---->	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
25	B	2	9.5	Non-Exposed / *Exposed	60 /*42	55 /*29	47 /*27	60 /*36	45 /*26	38 /*24	58 /*32	37 /*23	34 /*21	50 /*29	33 /*21	30 /*19	43 /*26	30 /*19	28 /*17	38 /*24	27 /*17	25 /*16	32 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7
		3	14.0	Non-Exposed / *Exposed	54 /*41	54 /*29	46 /*27	54 /*36	44 /*26	38 /*24	54 /*32	37 /*23	34 /*21	49 /*29	33 /*21	30 /*19	43 /*26	30 /*19	27 /*17	38 /*24	27 /*17	25 /*16	31 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7
		4	18.4	Non-Exposed / *Exposed	53 /*41	52 /*29	45 /*26	53 /*36	43 /*26	37 /*24	53 /*32	36 /*23	33 /*21	48 /*29	33 /*21	30 /*19	42 /*26	30 /*19	27 /*17	37 /*24	27 /*17	25 /*16	31 /*20	23 /*14	21 /*13	26 /*17	19 /*12	18 /*11	23 /*14	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	11 /*7	15 /*10	11 /*7	10 /*7
		5	22.6	Non-Exposed / *Exposed	53	53 /*35	NA	53	53 /*31	NA	53 /*45	49 /*28	NA	53 /*38	41 /*25	NA	53 /*34	36 /*23	NA	53 /*31	33 /*21	NA	43 /*26	27 /*18	NA	35 /*22	23 /*15	NA	30 /*19	20 /*13	NA	26 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA
		6	26.6	Non-Exposed / *Exposed	55	55 /*35	NA	55 /*54	55 /*31	NA	55 /*44	48 /*28	NA	55 /*37	40 /*25	NA	55 /*34	36 /*23	NA	55 /*31	32 /*21	NA	42 /*26	27 /*17	NA	34 /*22	23 /*15	NA	29 /*19	20 /*13	NA	25 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA
		8	33.7	Non-Exposed / *Exposed	59 /*54	43 /*24	35 /*18	59 /*34	35 /*21	29 /*16	59 /*37	31 /*19	25 /*15	59 /*34	27 /*17	21 /*13	57 /*31	24 /*16	18 /*12	48 /*28	27 /*14	17 /*11	36 /*23	19 /*12	14 /*9	31 /*20	16 /*10	12 /*8	27 /*17	14 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*12	9 /*6	7 /*4
		10	39.8	Non-Exposed / *Exposed	59 /*52	41 /*23	34 /*18	59 /*43	35 /*21	28 /*16	59 /*37	30 /*19	24 /*14	59 /*33	26 /*17	21 /*13	54 /*30	24 /*15	18 /*12	46 /*27	22 /*14	17 /*11	36 /*23	18 /*12	14 /*9	30 /*20	16 /*10	12 /*8	26 /*17	14 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*12	9 /*6	7 /*4
		12	45.0	Non-Exposed / *Exposed	57 /*49	40 /*23	33 /*18	57 /*41	34 /*21	28 /*16	57 /*36	29 /*19	24 /*14	57 /*33	26 /*17	20 /*13	52 /*30	24 /*15	18 /*12	44 /*27	22 /*14	17 /*11	35 /*23	18 /*12	14 /*9	30 /*20	16 /*10	12 /*8	26 /*17	13 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*12	9 /*6	7 /*4
	C	2	9.5	Non-Exposed / *Exposed	58 /*33	38 /*24	34 /*22	50 /*29	33 /*21	30 /*19	43 /*26	30 /*19	27 /*17	37 /*23	27 /*17	24 /*16	34 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	25 /*16	18 /*12	17 /*11	21 /*14	16 /*10	14 /*9	18 /*12	13 /*9	12 /*8	16 /*10	12 /*7	11 /*7	14 /*9	10 /*7	9 /*6	12 /*8	9 /*6	8 /*5
		3	14.0	Non-Exposed / *Exposed	54 /*32	37 /*23	34 /*21	49 /*29	33 /*21	30 /*19	42 /*26	29 /*19	27 /*17	37 /*23	27 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	25 /*16	18 /*12	17 /*11	21 /*14	16 /*10	14 /*9	18 /*12	13 /*9	12 /*8	16 /*10	12 /*7	11 /*7	14 /*9	10 /*7	9 /*6	12 /*8	9 /*6	8 /*5
		4	18.4	Non-Exposed / *Exposed	53 /*32	37 /*23	33 /*21	48 /*29	33 /*21	30 /*19	41 /*26	29 /*19	27 /*17	36 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	25 /*16	18 /*12	17 /*11	21 /*14	15 /*10	14 /*9	18 /*12	13 /*9	12 /*8	16 /*10	12 /*7	11 /*7	14 /*9	10 /*7	9 /*6	12 /*8	9 /*6	8 /*5
		5	22.6	Non-Exposed / *Exposed	53 /*46	50 /*28	NA	53 /*38	41 /*25	NA	53 /*34	35 /*23	NA	53 /*30	32 /*20	NA	48 /*27	29 /*19	NA	41 /*25	26 /*17	NA	33 /*21	22 /*14	NA	28 /*18	19 /*12	NA	24 /*15	16 /*10	NA	21 /*13	14 /*9	NA	18 /*12	12 /*8	NA	16 /*10	11 /*7	NA
		6	26.6	Non-Exposed / *Exposed	55 /*45	49 /*28	NA	55 /*37	40 /*25	NA	55 /*33	35 /*22	NA	55 /*30	32 /*20	NA	47 /*27	29 /*18	NA	40 /*25	26 /*17	NA	32 /*21	22 /*14	NA	28 /*18	19 /*12	NA	24 /*15	16 /*10	NA	21 /*13	14 /*9	NA	18 /*12	12 /*8	NA	16 /*10	11 /*7	NA
		8	33.7	Non-Exposed / *Exposed	59 /*38	31 /*19	25 /*15	59 /*34	26 /*17	21 /*13	55 /*30	24 /*15	18 /*12	46 /*27	22 /*14	16 /*11	39 /*25	20 /*13	15 /*10	35 /*23	18 /*12	14 /*9	30 /*19	15 /*10	12 /*7	25 /*16	13 /*8	10 /*6	22 /*14	13 /*7	8 /*5	19 /*12	10 /*6	7 /*5	17 /*11	8 /*5	6 /*4	15 /*9	7 /*5	6 /*4
		10	39.8	Non-Exposed / *Exposed	59 /*37	30 /*19	24 /*14	59 /*33	26 /*17	20 /*13	53 /*30	24 /*15	18 /*12	44 /*27	21 /*14	16 /*11	38 /*24	19 /*13	15 /*10	34 /*22	18 /*12	14 /*9	29 /*19	15 /*10	11 /*7	25 /*16	13 /*8	10 /*6	21 /*14	11 /*7	8 /*5	19 /*12	10 /*6	7 /*5	16 /*11	8 /*5	6 /*4	15 /*9	7 /*5	6 /*4
		12	45.0	Non-Exposed / *Exposed	57 /*36	30 /*19	24 /*14	57 /*32	26 /*17	20 /*13	50 /*29	23 /*15	18 /*12	43 /*27	21 /*14	16 /*11	37 /*24	19 /*13	15 /*10	34 /*22	18 /*11	13 /*9	29 /*19	15 /*10	11 /*7	24 /*16	13 /*8	10 /*6	21 /*14	11 /*7	8 /*5	18 /*12	10 /*6	7 /*5	16 /*11	8 /*5	6 /*4	14 /*9	7 /*5	6 /*4
30	D	2	9.5	Non-Exposed / *Exposed	43 /*26	30 /*19	27 /*17	37 /*23	27 /*17	24 /*16	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	27 /*17	20 /*12	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4
		3	14.0	Non-Exposed / *Exposed	42 /*26	30 /*19	27 /*17	37 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	21 /*14	20 /*13	27 /*17	19 /*12	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4
		4	18.4	Non-Exposed / *Exposed	42 /*26	29 /*19	27 /*17	36 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	29 /*19	21 /*14	19 /*13	27 /*17	19 /*12	18 /*11	24 /*15	18 /*11	16 /*10	20 /*13	15 /*9	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4
		5	22.6	Non-Exposed / *Exposed	53 /*34	36 /*23	NA	53 /*30	32 /*20	NA	47 /*27	29 /*18	NA	39 /*24	26 /*17	NA	35 /*22	23 /*15	NA	32 /*20	21 /*14	NA	26 /*17	18 /*12	NA	22 /*14	15 /*10	NA	19 /*12	13 /*8	NA	17 /*11	11 /*7	NA	15 /*9	10 /*6	NA	13 /*8	9 /*6	NA
		6	26.6	Non-Exposed / *Exposed	55 /*34	35 /*23	NA	55 /*30	32 /*20	NA	46 /*27	28 /*18	NA	38 /*24	26 /*16	NA	34 /*22	23 /*15	NA	31 /*20	21 /*14	NA	26 /*17	18 /*11	NA	22 /*14	15 /*10	NA	19 /*12	13 /*8	NA	17 /*11	11 /*7	NA	15 /*9	10 /*6	NA	13 /*8	9 /*6	NA
		8	33.7	Non-Exposed / *Exposed	56 /*30	24 /*16	18 /*12	46 /*27	22 /*14	16 /*11	38 /*25	19 /*13	15 /*10	34 /*22	18 /*11	13 /*9	31 /*20	16 /*10	12 /*8	28 /*18	15 /*9	11 /*7	24 /*16	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3
		10	39.8	Non-Exposed / *Exposed	53 /*30	24 /*15	18 /*12	44 /*27	21 /*14	16 /*11	37 /*24	19 /*12	15 /*10	34 /*22	17 /*11	13 /*9	31 /*20	16 /*10	12 /*8	28 /*18	14 /*9	11 /*7	24 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	17 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3
		12	45.0	Non-Exposed / *Exposed	51 /*30	23 /*15	18 /*12	43 /*26	21 /*14	16 /*11	37 /*24	19 /*12	14 /*9	34 /*22	17 /*11	13 /*9	30 /*20	16 /*10	12 /*8	28 /*18	14 /*9	11 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	17 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3
35	B	2	9.5	Non-Exposed / *Exposed	51 /*42	51 /*29	47 /*27	51 /*36	45 /*26	38 /*24	51 /*32	37 /*23	34 /*21	50 /*29	33 /*21	30 /*19	43 /*26	30 /*19	28 /*17	38 /*24	27 /*17	25 /*16	32 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7
		3	14.0	Non-Exposed / *Exposed	46 /*41	46 /*29	46 /*27	46 /*36	44 /*26	38 /*24	46 /*32	37 /*23	34 /*21	46 /*29	33 /*21	30 /*19	43 /*26	30 /*19	27 /*17	38 /*24	27 /*17	25 /*16	31 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7
		4	18.4	Non-Exposed / *Exposed	45 /*41	45 /*29	45 /*26	45 /*36	43 /*26	37 /*24	45 /*32	36 /*23	33 /*21	45 /*29	33 /*21	30 /*19	42 /*26	30 /*19	27 /*17	37 /*24	27 /*17	25 /*16	31 /*20	23 /*14	21 /*13	26 /*17	19 /*12	18 /*11	23 /*14	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	11 /*7	15 /*10	11 /*7	10 /*7
		5	22.6	Non-Exposed / *Exposed	46	46 /*35	NA	46	46 /*31	NA	46 /*45	46 /*28	NA	46 /*38	41 /*25	NA	46 /*34	36 /*23	NA	46 /*31	33 /*21	NA	43 /*26	27 /*18	NA	35 /*22	23 /*15	NA	30 /*19	20 /*13	NA	26 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA
		6	26.6	Non-Exposed / *Exposed	47	47 /*35	NA	47	47 /*31	NA	47 /*44	47 /*28	NA	47 /*37	40 /*25	NA	47 /*34	36 /*23	NA	47 /*31	32 /*21	NA	42 /*26	27 /*17	NA	34 /*22	23 /*15	NA	29 /*19	20 /*13	NA	25 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA
		8	33.7	Non-Exposed / *Exposed	53																																			

Roof Tech - Mini Gen II - ASCE 7-16 Hip - Flush-Mount - 15 Foot Max Mean Roof Height - Plywood Only - Portrait 72 Cell - 60 mm Long Screws																																											
Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->	95			100			105			110			115			120			130			140			150			160			170			180					
				Roof Area ---->	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3			
40	B	2	9.5	Non-Exposed / *Exposed	40	40 /*29	40 /*27	40 /*36	40 /*26	38 /*24	40 /*32	37 /*23	34 /*21	40 /*29	33 /*21	30 /*19	40 /*26	30 /*19	28 /*17	38 /*24	27 /*17	25 /*16	32 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7			
		3	14.0	Non-Exposed / *Exposed	36	36 /*29	36 /*27	36	36 /*26	36 /*24	36 /*32	36 /*23	34 /*21	36 /*29	33 /*21	30 /*19	36 /*26	30 /*19	27 /*17	36 /*24	27 /*17	25 /*16	31 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7			
		4	18.4	Non-Exposed / *Exposed	35	35 /*29	35 /*26	35	35 /*26	35 /*24	35 /*32	35 /*23	33 /*21	35 /*29	33 /*21	30 /*19	35 /*26	30 /*19	27 /*17	35 /*24	27 /*17	25 /*16	31 /*20	23 /*14	21 /*13	26 /*17	19 /*12	18 /*11	23 /*14	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	11 /*7	15 /*10	11 /*7	10 /*7			
		5	22.6	Non-Exposed / *Exposed	36	36 /*35	NA	36	36 /*31	NA	36	36 /*28	NA	36	36 /*25	NA	36	36 /*34	36 /*23	NA	36 /*31	33 /*21	NA	36 /*26	27 /*18	NA	35 /*22	23 /*15	NA	30 /*19	20 /*13	NA	26 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA		
		6	26.6	Non-Exposed / *Exposed	37	37 /*35	NA	37	37 /*31	NA	37	37 /*28	NA	37	37 /*25	NA	37	37 /*34	36 /*23	NA	37 /*31	32 /*21	NA	37 /*26	27 /*17	NA	34 /*22	23 /*15	NA	29 /*19	20 /*13	NA	25 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA		
		8	33.7	Non-Exposed / *Exposed	42	42 /*24	35 /*18	42	35 /*21	29 /*16	42 /*37	31 /*19	25 /*15	42 /*34	27 /*17	21 /*13	42 /*31	24 /*16	18 /*12	42 /*28	29 /*20	16 /*10	17 /*11	36 /*23	19 /*12	14 /*9	31 /*20	16 /*10	12 /*8	27 /*17	14 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*12	9 /*6	7 /*4		
		10	39.8	Non-Exposed / *Exposed	48	41 /*23	34 /*18	48 /*43	35 /*21	28 /*16	48 /*37	30 /*19	24 /*14	48 /*33	26 /*17	21 /*13	48 /*30	24 /*15	18 /*12	46 /*27	22 /*14	17 /*11	36 /*23	18 /*12	14 /*9	30 /*20	16 /*10	12 /*8	26 /*17	14 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*12	9 /*6	7 /*4			
		12	45.0	Non-Exposed / *Exposed	56 /*49	40 /*23	33 /*18	56 /*41	34 /*21	28 /*16	56 /*36	29 /*19	24 /*14	56 /*33	26 /*17	20 /*13	52 /*30	24 /*15	18 /*12	44 /*27	22 /*14	17 /*11	35 /*23	18 /*12	14 /*9	30 /*20	16 /*10	12 /*8	26 /*17	13 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*12	9 /*6	7 /*4			
	C	2	9.5	Non-Exposed / *Exposed	40 /*33	38 /*24	34 /*22	40 /*29	33 /*21	30 /*19	40 /*26	30 /*19	27 /*17	37 /*23	27 /*17	24 /*16	34 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	25 /*16	18 /*12	17 /*11	21 /*14	16 /*10	14 /*9	18 /*12	13 /*9	12 /*8	16 /*10	12 /*7	11 /*7	14 /*9	10 /*7	9 /*6	12 /*8	9 /*6	8 /*5			
		3	14.0	Non-Exposed / *Exposed	36 /*32	36 /*23	34 /*21	36 /*29	33 /*21	30 /*19	36 /*26	29 /*19	27 /*17	36 /*23	27 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	25 /*16	18 /*12	17 /*11	21 /*14	16 /*10	14 /*9	18 /*12	13 /*9	12 /*8	16 /*10	12 /*7	11 /*7	14 /*9	10 /*7	9 /*6	12 /*8	9 /*6	8 /*5			
		4	18.4	Non-Exposed / *Exposed	35 /*32	35 /*23	33 /*21	35 /*29	33 /*21	30 /*19	35 /*26	29 /*19	27 /*17	35 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	25 /*16	18 /*12	17 /*11	21 /*14	15 /*10	14 /*9	18 /*12	13 /*9	12 /*8	16 /*10	12 /*7	11 /*7	14 /*9	10 /*7	9 /*6	12 /*8	9 /*6	8 /*5			
		5	22.6	Non-Exposed / *Exposed	36	36 /*28	NA	36	36 /*25	NA	36	36 /*34	35 /*23	NA	36	36 /*30	32 /*20	NA	36 /*27	29 /*18	NA	36 /*25	26 /*17	NA	33 /*21	22 /*14	NA	28 /*18	19 /*12	NA	24 /*15	16 /*10	NA	21 /*13	14 /*9	NA	18 /*12	12 /*8	NA	16 /*10	11 /*7	NA	
		6	26.6	Non-Exposed / *Exposed	37	37 /*28	NA	37	37 /*25	NA	37	37 /*33	35 /*22	NA	37	37 /*30	32 /*20	NA	37 /*27	29 /*18	NA	37 /*25	26 /*17	NA	32 /*21	22 /*14	NA	28 /*18	19 /*12	NA	24 /*15	16 /*10	NA	21 /*13	14 /*9	NA	18 /*12	12 /*8	NA	16 /*10	11 /*7	NA	
		8	33.7	Non-Exposed / *Exposed	42 /*38	31 /*19	25 /*15	42 /*34	26 /*17	21 /*13	42 /*30	24 /*15	18 /*12	42 /*27	22 /*14	16 /*11	39 /*25	20 /*13	15 /*10	35 /*23	18 /*12	14 /*9	30 /*19	22 /*14	15 /*10	12 /*7	25 /*16	13 /*8	10 /*6	22 /*14	11 /*7	8 /*5	19 /*12	10 /*6	7 /*5	17 /*11	8 /*5	6 /*4	15 /*9	7 /*5	6 /*4		
		10	39.8	Non-Exposed / *Exposed	48 /*37	30 /*19	24 /*14	48 /*33	26 /*17	20 /*13	48 /*30	24 /*15	18 /*12	44 /*27	21 /*14	16 /*11	38 /*24	19 /*13	15 /*10	34 /*22	18 /*12	14 /*9	29 /*19	15 /*10	11 /*7	25 /*16	13 /*8	10 /*6	21 /*14	11 /*7	8 /*5	19 /*12	10 /*6	7 /*5	16 /*11	8 /*5	6 /*4	15 /*9	7 /*5	6 /*4			
		12	45.0	Non-Exposed / *Exposed	56 /*36	30 /*19	24 /*14	56 /*32	26 /*17	20 /*13	50 /*29	23 /*15	18 /*12	43 /*27	21 /*14	16 /*11	37 /*24	19 /*13	15 /*10	34 /*22	18 /*11	13 /*9	29 /*19	15 /*10	11 /*7	24 /*16	13 /*8	10 /*6	21 /*14	11 /*7	8 /*5	18 /*12	10 /*6	7 /*5	16 /*11	8 /*5	6 /*4	14 /*9	7 /*5	6 /*4			
	D	2	9.5	Non-Exposed / *Exposed	40 /*26	30 /*19	27 /*17	37 /*23	27 /*17	24 /*16	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	27 /*17	20 /*12	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4			
		3	14.0	Non-Exposed / *Exposed	36 /*26	30 /*19	27 /*17	36 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	21 /*14	20 /*13	27 /*17	19 /*12	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4			
		4	18.4	Non-Exposed / *Exposed	35 /*26	29 /*19	27 /*17	35 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	29 /*19	21 /*14	19 /*13	27 /*17	19 /*12	18 /*11	24 /*15	18 /*11	16 /*10	20 /*13	15 /*9	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4			
		5	22.6	Non-Exposed / *Exposed	36 /*34	36 /*23	NA	36	36 /*30	32 /*20	NA	36	36 /*27	29 /*18	NA	36	36 /*24	26 /*17	NA	35 /*22	23 /*15	NA	32 /*20	21 /*14	NA	26 /*17	18 /*12	NA	22 /*14	15 /*10	NA	19 /*12	13 /*8	NA	17 /*11	11 /*7	NA	15 /*9	10 /*6	NA	13 /*8	9 /*6	NA
		6	26.6	Non-Exposed / *Exposed	37 /*34	35 /*23	NA	37	37 /*30	32 /*20	NA	37	37 /*27	28 /*18	NA	37	37 /*24	26 /*16	NA	34 /*22	23 /*15	NA	31 /*20	21 /*14	NA	26 /*17	18 /*11	NA	22 /*14	15 /*10	NA	19 /*12	13 /*8	NA	17 /*11	11 /*7	NA	15 /*9	10 /*6	NA	13 /*8	9 /*6	NA
		8	33.7	Non-Exposed / *Exposed	42 /*30	24 /*16	18 /*12	42 /*27	22 /*14	16 /*11	38 /*25	19 /*13	15 /*10	34 /*22	18 /*11	13 /*9	31 /*20	16 /*10	12 /*8	28 /*18	15 /*9	11 /*7	24 /*16	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3			
		10	39.8	Non-Exposed / *Exposed	48 /*30	24 /*15	18 /*12	44 /*27	21 /*14	16 /*11	37 /*24	19 /*12	15 /*10	34 /*22	17 /*11	13 /*9	31 /*20	16 /*10	12 /*8	28 /*18	14 /*9	11 /*7	24 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	17 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3			
		12	45.0	Non-Exposed / *Exposed	51 /*30	23 /*15	18 /*12	43 /*26	21 /*14	16 /*11	37 /*24	19 /*12	15 /*9	33 /*22	17 /*11	13 /*9	30 /*20	16 /*10	12 /*8	28 /*18	14 /*9	11 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	17 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3			
50	B	2	9.5	Non-Exposed / *Exposed	32	32 /*29	32 /*27	32	32 /*26	32 /*24	32	32 /*23	32 /*21	32 /*29	32 /*21	30 /*19	32 /*26	30 /*19	28 /*17	32 /*24	27 /*17	25 /*16	32 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7			
		3	14.0	Non-Exposed / *Exposed	29	29	29 /*27	29	29 /*26	29 /*24	29	29 /*23	29 /*21	29	29 /*21	29 /*19	29 /*26	29 /*19	27 /*17	29 /*24	27 /*17	25 /*16	29 /*20	23 /*15	21 /*13	27 /*17	19 /*12	18 /*11	23 /*15	17 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*7			
		4	18.4	Non-Exposed / *Exposed	29	29	29 /*26	29	29 /*26	29 /*24	29	29 /*23	29 /*21	29	29 /*21	29 /*19	29 /*26	29 /*19	27 /*17	29 /*24	27 /*17	25 /*16	29 /*20	23 /*14	21 /*13	26 /*17	19 /*12	18 /*11	23 /*14	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	11 /*7	15 /*10	11 /*7	10 /*7			
		5	22.6	Non-Exposed / *Exposed	29	29	NA	29	NA	NA	29	29 /*28	NA	29	29 /*25	NA	29	29 /*23	NA	29	29 /*21	NA	29	29 /*22	23 /*15	NA	29 /*22	23 /*15	NA	29 /*19	20 /*13	NA	26 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA		
		6	26.6	Non-Exposed / *Exposed	30	30	NA	30	NA	NA	30	30 /*28	NA	30	30 /*25	NA	30	30 /*23	NA	30	30 /*21	NA	30	30 /*26	27 /*17	NA	30 /*22	23 /*15	NA	29 /*19	20 /*13	NA	25 /*16	17 /*11	NA	22 /*14	15 /*10	NA	20 /*13	13 /*9	NA		
		8	33.7	Non-Exposed / *Exposed	35	35 /*																																					

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

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Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

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Roof Tech - Mini Gen II - ASCE 7-16 Gable - Flush-Mount - 30 Foot Max Mean Roof Height - Rafter Offset - Portrait 60 Cell - (5) 60 mm Long Screws

Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->			95			100			105			110			115			120			130			140			150			160			170			180				
				Roof Area ---->			1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3					
							1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3					
B	0	0.0	Non-Exposed / *Exposed	96	96	96/*68	96	96	96/*57	96	96/*82	96/*48	96	96/*70	84/*41	96	96/*60	71/*38	96/*88	96/*53	61/*34	96/*67	79/*41	46/*29	96/*52	62/*35	38/*25	82/*42	49/*30	33/*21	66/*36	41/*26	29/*19	54/*32	36/*23	25/*16	45/*28	32/*21	22/*14					
	1	4.8	Non-Exposed / *Exposed	96	96	96/*67	96	96	96/*56	96	96/*81	96/*48	96	96/*69	83/*41	96	96/*60	70/*37	96/*87	96/*52	60/*34	96/*66	78/*41	46/*29	96/*52	61/*35	38/*25	81/*42	49/*30	33/*21	65/*36	41/*26	28/*19	53/*32	36/*23	25/*16	44/*28	32/*20	22/*14					
	2	9.5	Non-Exposed / *Exposed	96	96/*81	96/*56	96	96	96/*66	96/*46	96	96/*54	87/*40	96/*77	96/*45	71/*36	96/*68	86/*40	59/*33	96/*61	71/*36	50/*30	90/*49	52/*31	39/*25	69/*40	40/*26	33/*22	57/*35	35/*23	29/*19	48/*30	30/*20	25/*16	41/*27	27/*17	22/*14	36/*24	24/*15	20/*13				
	3	14.0	Non-Exposed / *Exposed	96	96/*80	96/*56	96	96	96/*64	96/*46	96/*95	96/*53	85/*40	96/*76	96/*45	70/*36	96/*67	84/*40	58/*33	96/*60	70/*36	49/*30	85/*48	51/*31	39/*25	69/*40	40/*26	33/*22	57/*35	35/*23	29/*19	48/*30	30/*20	25/*16	41/*27	27/*17	22/*14	36/*24	24/*15	19/*13				
	4	18.4	Non-Exposed / *Exposed	96	96/*78	96/*55	96	96	96/*63	96/*45	96/*90	96/*53	83/*39	96/*75	96/*44	68/*36	96/*67	82/*39	57/*33	96/*59	69/*36	49/*30	83/*48	51/*30	39/*25	68/*40	40/*26	33/*22	56/*35	35/*23	28/*19	47/*30	30/*20	25/*16	41/*27	27/*17	22/*14	36/*24	24/*15	19/*13				
	5	22.6	Non-Exposed / *Exposed	96	96	96/*88	96	96	96/*88	96/*67	96	96/*72	96/*55	96	96/*61	96/*46	96	96/*52	89/*39	96/*88	96/*45	73/*35	96/*70	69/*37	52/*28	96/*57	53/*31	40/*23	82/*48	42/*27	33/*19	69/*41	36/*24	28/*16	59/*36	32/*21	24/*14	50/*32	28/*19	20/*13				
	6	26.6	Non-Exposed / *Exposed	96	96	96/*82	96	96	96/*86	96/*66	96	96/*71	96/*54	96	96/*60	96/*45	96	96/*51	87/*39	96/*86	93/*44	72/*35	96/*69	68/*37	52/*28	96/*57	52/*31	40/*23	81/*47	41/*27	33/*19	68/*40	36/*24	28/*16	58/*36	32/*21	23/*14	50/*32	28/*19	20/*13				
	8	33.7	Non-Exposed / *Exposed	96	96	96	96	96	96/*81	96	96	96	96/*68	96	96	96/*85	96/*58	96	96/*72	96/*50	96/*83	96/*62	87/*43	96/*60	66/*35	96/*45	74/*39	51/*29	76/*38	58/*34	41/*25	59/*33	47/*30	35/*21	47/*29	40/*26	30/*18	39/*26	35/*23	26/*16				
	10	39.8	Non-Exposed / *Exposed	96	96	96/*94	96	96	96/*79	96	96	96	96/*66	96	96	96/*82	96/*57	96	96/*70	96/*49	96/*80	96/*61	85/*43	96/*58	93/*47	64/*35	96/*44	72/*39	50/*29	74/*37	57/*34	40/*24	57/*33	46/*29	34/*21	46/*29	39/*26	30/*18	39/*26	35/*23	26/*16			
	12	45.0	Non-Exposed / *Exposed	96	96	96/*92	96	96	96/*77	96	96	96/*93	96/*65	96	96/*80	96/*56	96/*92	96/*69	95/*48	96/*78	96/*60	82/*42	96/*57	90/*46	63/*35	95/*44	70/*38	49/*29	72/*37	56/*33	40/*24	56/*33	46/*29	34/*21	45/*29	39/*26	29/*18	39/*26	35/*23	26/*16				
	C	0	0.0	Non-Exposed / *Exposed	96	96/*65	77/*39	96/*93	96/*55	64/*35	96/*78	93/*47	54/*32	96/*66	79/*41	46/*29	96/*57	68/*37	40/*26	96/*50	59/*34	37/*24	75/*40	45/*29	31/*20	58/*34	38/*24	27/*17	47/*29	33/*21	23/*15	39/*25	28/*18	20/*13	34/*22	25/*16	18/*11	30/*20	22/*14	16/*10				
		1	4.8	Non-Exposed / *Exposed	96	96/*64	76/*39	96/*91	96/*54	63/*35	96/*77	91/*47	53/*32	96/*65	77/*41	46/*29	96/*57	67/*37	40/*26	96/*49	58/*34	37/*24	74/*39	45/*29	31/*20	58/*34	38/*24	26/*17	46/*29	32/*21	23/*15	39/*25	28/*18	20/*13	34/*22	25/*16	18/*11	30/*20	22/*14	16/*10				
2		9.5	Non-Exposed / *Exposed	96/*73	95/*42	65/*34	96/*63	75/*37	53/*31	96/*55	62/*34	44/*28	88/*49	51/*31	39/*25	75/*43	44/*28	35/*23	67/*39	39/*25	32/*21	53/*33	33/*21	27/*18	44/*28	28/*18	23/*15	37/*24	24/*16	20/*13	33/*21	21/*14	18/*11	29/*19	19/*12	15/*10	25/*17	17/*11	14/*9					
3		14.0	Non-Exposed / *Exposed	96/*72	93/*42	64/*34	96/*62	74/*37	52/*31	96/*55	61/*34	43/*28	84/*48	51/*30	39/*25	74/*43	43/*28	35/*23	66/*39	39/*25	32/*21	53/*33	33/*21	27/*18	44/*28	28/*18	23/*15	37/*24	24/*16	20/*13	33/*21	21/*14	17/*11	29/*19	19/*12	15/*10	25/*17	17/*11	14/*9					
4		18.4	Non-Exposed / *Exposed	96/*71	90/*41	62/*34	96/*62	73/*37	51/*31	96/*54	60/*33	43/*28	83/*48	50/*30	38/*25	73/*43	43/*28	35/*23	65/*39	39/*25	32/*21	52/*33	33/*21	27/*18	43/*28	28/*18	23/*15	37/*24	24/*16	20/*13	32/*21	21/*14	17/*11	28/*19	19/*12	15/*10	25/*16	16/*11	14/*9					
5		22.6	Non-Exposed / *Exposed	96	96/*56	96/*42	96/*93	96/*47	78/*36	96/*79	83/*40	63/*32	96/*69	69/*37	52/*28	96/*62	58/*33	44/*25	96/*55	50/*30	38/*22	77/*45	39/*26	31/*18	63/*38	34/*22	25/*15	52/*33	29/*19	21/*13	44/*28	25/*17	18/*11	38/*25	22/*15	15/*10	34/*22	20/*13	14/*9					
6		26.6	Non-Exposed / *Exposed	96	96/*55	96/*42	96/*91	96/*46	76/*36	96/*78	81/*40	62/*32	96/*69	68/*36	51/*28	96/*61	58/*33	43/*25	96/*54	50/*30	38/*22	76/*44	39/*26	31/*18	62/*37	34/*22	25/*15	52/*32	29/*19	21/*13	44/*28	25/*17	18/*11	38/*25	22/*15	15/*10	34/*22	20/*13	14/*9					
8		33.7	Non-Exposed / *Exposed	96	96/*78	96/*54	96/*88	96/*65	92/*45	96/*71	96/*55	77/*39	96/*59	96/*48	65/*35	96/*50	81/*41	56/*31	96/*43	70/*38	49/*28	68/*36	53/*32	38/*23	51/*31	42/*28	32/*19	41/*27	36/*24	27/*16	35/*23	32/*21	23/*14	31/*21	28/*18	20/*12	28/*18	25/*16	17/*10					
10		39.8	Non-Exposed / *Exposed	96	96/*76	96/*53	96/*85	96/*64	89/*45	96/*69	96/*54	75/*39	96/*58	92/*47	64/*35	96/*49	79/*41	55/*31	92/*42	68/*38	48/*28	66/*36	52/*32	38/*23	50/*30	42/*27	32/*19	40/*26	36/*24	27/*16	35/*23	32/*21	23/*14	31/*20	28/*18	20/*12	28/*18	25/*16	17/*10					
12		45.0	Non-Exposed / *Exposed	96	96/*74	96/*52	96/*82	96/*63	86/*44	96/*67	96/*53	73/*39	96/*57	89/*46	62/*34	96/*48	77/*41	54/*31	89/*42	67/*37	47/*28	65/*35	52/*32	38/*23	49/*30	41/*27	31/*19	40/*26	36/*24	26/*16	35/*23	31/*21	23/*14	31/*20	28/*18	19/*12	27/*18	25/*16	17/*10					

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

Roof Tech - Mini Gen II - ASCE 7-16 Gable - Flush-Mount - 30 Foot Max Mean Roof Height - Plywood Only - Portrait 72 Cell - 60 mm Long Screws																																								
Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->	95			100			105			110			115			120			130			140			150			160			170			180		
				Roof Area ---->			1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3				
		0	0.0	Non-Exposed / *Exposed	32	32 /*31	32 /*21	32	32 /*28	30 /*19	32	32 /*25	27 /*17	32	32 /*22	24 /*15	32 /*29	32 /*20	22 /*14	32 /*26	29 /*18	20 /*13	32 /*22	24 /*15	17 /*11	29 /*18	21 /*13	14 /*9	25 /*16	18 /*11	12 /*8	21 /*14	15 /*10	11 /*7	19 /*12	13 /*9	9 /*6	16 /*10	12 /*8	8 /*5
		1	4.8	Non-Exposed / *Exposed	27	27	27 /*21	27	27	27 /*19	27	27 /*25	27 /*17	27	27 /*22	24 /*15	27	27 /*20	22 /*14	27 /*26	27 /*18	20 /*13	27 /*21	24 /*15	17 /*11	27 /*18	20 /*13	14 /*9	25 /*16	18 /*11	12 /*8	21 /*14	15 /*10	11 /*7	19 /*12	13 /*8	9 /*6	16 /*10	12 /*7	8 /*5
		2	9.5	Non-Exposed / *Exposed	24	24 /*23	24 /*19	24	24 /*20	24 /*17	24	24 /*18	23 /*15	24	24 /*16	21 /*13	24 /*23	23 /*15	19 /*12	24 /*21	21 /*14	17 /*11	24 /*18	18 /*11	15 /*9	24 /*15	15 /*10	12 /*8	20 /*13	13 /*8	11 /*7	18 /*11	11 /*7	9 /*6	15 /*10	10 /*6	8 /*5	14 /*9	9 /*6	7 /*5
		3	14.0	Non-Exposed / *Exposed	21	21	21 /*18	21	21 /*20	21 /*17	21	21 /*18	21 /*15	21	21 /*16	21 /*13	21	21 /*15	19 /*12	21	21 /*14	17 /*11	21 /*18	18 /*11	15 /*9	21 /*15	15 /*10	12 /*8	20 /*13	13 /*8	11 /*7	18 /*11	11 /*7	9 /*6	15 /*10	10 /*6	8 /*5	14 /*9	9 /*6	7 /*5
		4	18.4	Non-Exposed / *Exposed	21	21	21 /*18	21	21 /*20	21 /*16	21	21 /*18	21 /*15	21	21 /*16	21 /*13	21	21 /*15	19 /*12	21	21 /*13	17 /*11	21 /*18	18 /*11	14 /*9	21 /*15	15 /*10	12 /*8	20 /*13	13 /*8	11 /*7	17 /*11	11 /*7	9 /*6	15 /*10	10 /*6	8 /*5	13 /*9	9 /*6	7 /*5
		5	22.6	Non-Exposed / *Exposed	21	21	21 /*19	21	21	21 /*17	21	21	21 /*15	21	21 /*20	21 /*13	21	21 /*18	20 /*12	21	21 /*16	18 /*11	21	21 /*14	14 /*9	21 /*20	18 /*12	12 /*8	21 /*18	16 /*10	11 /*7	21 /*15	14 /*9	9 /*6	21 /*13	12 /*8	8 /*5	18 /*12	11 /*7	7 /*5
		6	26.6	Non-Exposed / *Exposed	22	22	22 /*19	22	22	22 /*17	22	22	22 /*15	22	22 /*20	22 /*13	22	22 /*18	20 /*12	22	22 /*16	18 /*11	22	21 /*14	14 /*9	22 /*20	18 /*12	12 /*8	22 /*17	16 /*10	10 /*7	22 /*15	14 /*9	9 /*6	21 /*13	12 /*8	8 /*5	18 /*12	10 /*7	7 /*5
		8	33.7	Non-Exposed / *Exposed	26	26	26 /*25	26	26	26 /*22	26	26	26 /*19	26	26 /*25	26 /*17	26 /*25	26 /*22	25 /*15	26 /*23	26 /*20	23 /*13	26 /*19	26 /*17	18 /*11	25 /*16	23 /*15	15 /*9	22 /*14	20 /*13	13 /*8	19 /*12	17 /*11	11 /*7	17 /*11	15 /*10	9 /*6	15 /*10	13 /*9	8 /*5
		10	39.8	Non-Exposed / *Exposed	30	30	30 /*24	30	30	30 /*21	30	30 /*27	30 /*19	30 /*27	30 /*24	28 /*17	30 /*25	30 /*22	25 /*15	30 /*23	30 /*20	22 /*13	29 /*19	26 /*17	18 /*11	25 /*16	22 /*15	15 /*9	22 /*14	19 /*13	13 /*8	19 /*12	17 /*11	11 /*7	17 /*11	15 /*10	9 /*6	15 /*10	13 /*9	8 /*5
		12	45.0	Non-Exposed / *Exposed	37	37 /*33	37 /*24	37 /*33	37 /*29	35 /*21	37 /*30	37 /*27	31 /*19	37 /*27	37 /*24	28 /*16	37 /*24	34 /*22	25 /*15	34 /*22	31 /*20	22 /*13	29 /*19	26 /*17	18 /*11	25 /*16	22 /*14	15 /*9	21 /*14	19 /*12	13 /*8	19 /*12	17 /*11	11 /*7	16 /*11					

[illegible]

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

[illegible]

[illegible]

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

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Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

Roof Tech - Mini Gen II - ASCE 7-16 Hip - Flush-Mount - 30 Foot Max Mean Roof Height - Plywood Only - Portrait 60 Cell - 60 mm Long Screws																																								
Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->	95			100			105			110			115			120			130			140			150			160			170			180		
				Roof Area ---->				1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3			
0	B	2	9.5	Non-Exposed / *Exposed	78 /*48	63 /*34	53 /*31	77 /*41	51 /*30	44 /*27	66 /*37	43 /*27	39 /*24	57 /*33	38 /*24	35 /*22	50 /*30	34 /*22	31 /*20	44 /*27	31 /*20	29 /*18	36 /*23	26 /*17	24 /*15	31 /*19	22 /*14	20 /*13	26 /*17	19 /*12	17 /*11	23 /*15	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*8
		3	14.0	Non-Exposed / *Exposed	76 /*47	61 /*33	52 /*30	75 /*41	50 /*30	43 /*27	64 /*37	42 /*27	38 /*24	56 /*33	38 /*24	34 /*22	49 /*30	34 /*22	31 /*20	43 /*27	31 /*20	28 /*18	36 /*23	26 /*17	24 /*15	30 /*19	22 /*14	20 /*13	26 /*17	19 /*12	17 /*11	22 /*14	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7
		4	18.4	Non-Exposed / *Exposed	75 /*46	60 /*33	51 /*30	73 /*41	49 /*29	42 /*27	63 /*36	42 /*26	38 /*24	55 /*33	37 /*24	34 /*22	48 /*30	34 /*22	31 /*20	43 /*27	31 /*20	28 /*18	35 /*23	26 /*17	24 /*15	30 /*19	22 /*14	20 /*13	26 /*17	19 /*12	17 /*11	22 /*14	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7
		5	22.6	Non-Exposed / *Exposed	73	73 /*40	NA	73 /*64	69 /*36	NA	73 /*52	57 /*32	NA	73 /*43	47 /*29	NA	73 /*39	41 /*26	NA	70 /*35	37 /*24	NA	50 /*30	31 /*20	NA	39 /*25	26 /*17	NA	34 /*22	23 /*15	NA	29 /*19	20 /*13	NA	26 /*16	17 /*11	NA	23 /*15	15 /*10	NA
		6	26.6	Non-Exposed / *Exposed	72	72 /*40	NA	72 /*62	67 /*35	NA	72 /*51	55 /*32	NA	72 /*43	46 /*29	NA	72 /*38	41 /*26	NA	67 /*35	37 /*24	NA	48 /*29	31 /*20	NA	39 /*25	26 /*17	NA	33 /*21	23 /*15	NA	29 /*19	20 /*13	NA	25 /*16	17 /*11	NA	22 /*15	15 /*10	NA
		8	33.7	Non-Exposed / *Exposed	69 /*62	50 /*27	40 /*21	69 /*51	41 /*24	33 /*18	69 /*43	35 /*22	28 /*17	69 /*38	30 /*20	24 /*15	65 /*35	28 /*18	21 /*14	55 /*32	25 /*16	19 /*12	42 /*27	21 /*14	16 /*11	35 /*23	18 /*12	14 /*9	30 /*20	16 /*10	12 /*8	27 /*17	14 /*9	10 /*7	23 /*15	12 /*8	9 /*6	21 /*13	11 /*7	8 /*5
	C	10	39.8	Non-Exposed / *Exposed	67 /*59	47 /*27	39 /*20	67 /*49	39 /*24	33 /*18	67 /*42	34 /*21	28 /*16	67 /*38	30 /*19	24 /*15	61 /*34	27 /*18	21 /*14	53 /*31	25 /*16	19 /*12	41 /*26	21 /*14	16 /*10	35 /*23	18 /*12	14 /*9	30 /*20	15 /*10	12 /*8	26 /*17	14 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	11 /*7	8 /*5
		12	45.0	Non-Exposed / *Exposed	65 /*57	46 /*26	38 /*20	65 /*47	39 /*24	32 /*18	65 /*41	34 /*21	27 /*16	65 /*37	30 /*19	23 /*15	59 /*34	27 /*18	21 /*13	51 /*31	25 /*16	19 /*12	40 /*26	21 /*14	16 /*10	34 /*22	18 /*12	14 /*9	30 /*19	15 /*10	12 /*8	26 /*17	13 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5
		2	9.5	Non-Exposed / *Exposed	53 /*32	36 /*23	33 /*21	46 /*28	32 /*20	29 /*19	40 /*25	29 /*18	26 /*17	36 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	21 /*14	20 /*13	25 /*16	18 /*12	16 /*11	21 /*13	15 /*10	14 /*9	18 /*12	13 /*8	12 /*8	16 /*10	11 /*7	10 /*7	14 /*9	10 /*6	9 /*6	12 /*8	9 /*6	8 /*5
		3	14.0	Non-Exposed / *Exposed	52 /*31	36 /*23	33 /*21	45 /*28	32 /*20	29 /*19	40 /*25	29 /*18	26 /*17	36 /*23	26 /*17	24 /*15	32 /*21	23 /*15	21 /*14	29 /*19	21 /*14	19 /*13	25 /*16	18 /*11	16 /*11	21 /*13	15 /*10	14 /*9	18 /*12	13 /*8	12 /*8	16 /*10	11 /*7	10 /*7	14 /*9	10 /*6	9 /*6	12 /*8	9 /*6	8 /*5
		4	18.4	Non-Exposed / *Exposed	51 /*31	35 /*23	32 /*21	44 /*28	32 /*20	29 /*19	39 /*25	28 /*18	26 /*17	35 /*23	26 /*16	23 /*15	32 /*20	23 /*15	21 /*14	29 /*19	21 /*14	19 /*12	24 /*16	18 /*11	16 /*11	21 /*13	15 /*10	14 /*9	18 /*11	13 /*8	12 /*8	15 /*10	11 /*7	10 /*7	14 /*9	10 /*6	9 /*6	12 /*8	9 /*6	8 /*5
		5	22.6	Non-Exposed / *Exposed	73 /*41	44 /*27	NA	73 /*36	38 /*25	NA	60 /*33	34 /*22	NA	50 /*29	31 /*20	NA	42 /*27	28 /*18	NA	38 /*24	26 /*17	NA	32 /*20	22 /*14	NA	27 /*17	18 /*12	NA	23 /*15	16 /*10	NA	20 /*13	14 /*9	NA	18 /*11	12 /*8	NA	16 /*10	11 /*7	NA
10	B	6	26.6	Non-Exposed / *Exposed	72 /*40	43 /*27	NA	72 /*36	38 /*24	NA	59 /*32	34 /*22	NA	49 /*29	31 /*20	NA	41 /*27	28 /*18	NA	38 /*24	25 /*16	NA	32 /*20	21 /*14	NA	27 /*17	18 /*12	NA	23 /*15	16 /*10	NA	20 /*13	14 /*9	NA	18 /*11	12 /*8	NA	16 /*10	11 /*7	NA
		8	33.7	Non-Exposed / *Exposed	69 /*37	29 /*19	23 /*14	58 /*33	26 /*17	20 /*13	49 /*30	23 /*15	18 /*12	41 /*27	21 /*14	16 /*10	38 /*24	19 /*12	15 /*10	34 /*22	18 /*11	13 /*9	29 /*19	15 /*10	11 /*7	25 /*16	13 /*8	10 /*6	21 /*14	11 /*7	8 /*5	18 /*12	9 /*6	7 /*5	16 /*11	8 /*5	6 /*4	14 /*9	7 /*5	6 /*4
		10	39.8	Non-Exposed / *Exposed	67 /*36	29 /*19	22 /*14	56 /*32	26 /*17	20 /*13	47 /*29	23 /*15	18 /*11	41 /*26	21 /*14	16 /*10	37 /*24	19 /*12	15 /*9	34 /*22	17 /*11	13 /*9	28 /*18	15 /*10	11 /*7	24 /*16	13 /*8	10 /*6	21 /*14	11 /*7	8 /*5	18 /*12	9 /*6	7 /*5	16 /*10	8 /*5	6 /*4	14 /*9	7 /*5	6 /*4
		12	45.0	Non-Exposed / *Exposed	64 /*35	28 /*18	22 /*14	54 /*32	25 /*17	19 /*13	45 /*29	23 /*15	18 /*11	40 /*26	21 /*14	16 /*10	36 /*24	19 /*12	14 /*9	33 /*22	17 /*11	13 /*9	28 /*18	15 /*9	11 /*7	24 /*16	12 /*8	10 /*6	21 /*14	11 /*7	8 /*5	18 /*12	9 /*6	7 /*5	16 /*10	8 /*5	6 /*4	14 /*9	7 /*5	6 /*4
		2	9.5	Non-Exposed / *Exposed	42 /*26	30 /*19	27 /*17	37 /*23	27 /*17	24 /*16	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	27 /*17	20 /*13	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4
		3	14.0	Non-Exposed / *Exposed	41 /*26	30 /*19	27 /*17	37 /*23	26 /*17	24 /*16	33 /*21	24 /*15	22 /*14	29 /*19	21 /*14	20 /*13	27 /*17	19 /*12	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4
	C	4	18.4	Non-Exposed / *Exposed	41 /*26	29 /*19	27 /*17	36 /*23	26 /*17	24 /*15	32 /*21	24 /*15	22 /*14	29 /*19	21 /*14	20 /*13	26 /*17	19 /*12	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4
		5	22.6	Non-Exposed / *Exposed	65 /*34	36 /*23	NA	52 /*30	32 /*20	NA	43 /*27	29 /*18	NA	38 /*24	26 /*17	NA	35 /*22	23 /*15	NA	32 /*20	21 /*14	NA	26 /*17	18 /*12	NA	23 /*15	15 /*10	NA	19 /*13	13 /*9	NA	17 /*11	11 /*7	NA	15 /*10	10 /*6	NA	13 /*8	9 /*6	NA
		6	26.6	Non-Exposed / *Exposed	63 /*34	35 /*23	NA	51 /*30	32 /*20	NA	42 /*27	28 /*18	NA	38 /*24	26 /*17	NA	34 /*22	23 /*15	NA	31 /*20	21 /*14	NA	26 /*17	18 /*12	NA	22 /*14	15 /*10	NA	19 /*12	13 /*8	NA	17 /*11	11 /*7	NA	15 /*10	10 /*6	NA	13 /*8	9 /*6	NA
		8	33.7	Non-Exposed / *Exposed	52 /*31	24 /*16	18 /*12	43 /*27	22 /*14	17 /*11	38 /*25	19 /*13	15 /*10	34 /*22	18 /*11	13 /*9	31 /*20	16 /*10	12 /*8	29 /*19	15 /*9	11 /*7	24 /*16	12 /*8	9 /*6	21 /*13	11 /*7	8 /*5	18 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	14 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3
		10	39.8	Non-Exposed / *Exposed	50 /*30	24 /*16	18 /*12	42 /*27	21 /*14	16 /*11	37 /*24	19 /*13	15 /*10	34 /*22	17 /*11	13 /*9	31 /*20	16 /*10	12 /*8	28 /*18	15 /*9	11 /*7	24 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3
		12	45.0	Non-Exposed / *Exposed	48 /*30	24 /*15	18 /*12	41 /*27	21 /*14	16 /*11	37 /*24	19 /*12	15 /*10	33 /*22	17 /*11	13 /*9	30 /*20	16 /*10	12 /*8	28 /*18	14 /*9	11 /*7	24 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	17 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9					

Roof Tech - Mini Gen II - ASCE 7-16 Hip - Flush-Mount - 30 Foot Max Mean Roof Height - Plywood Only - Portrait 60 Cell - 60 mm Long Screws																																								
Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->	95			100			105			110			115			120			130			140			150			160			170			180		
				Roof Area ---->				1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3			
25	B	2	9.5	Non-Exposed / *Exposed	68 /*48	63 /*34	53 /*31	68 /*41	51 /*30	44 /*27	66 /*37	43 /*27	39 /*24	57 /*33	38 /*24	35 /*22	50 /*30	34 /*22	31 /*20	44 /*27	31 /*20	29 /*18	36 /*23	26 /*17	24 /*15	31 /*19	22 /*14	20 /*13	26 /*17	19 /*12	17 /*11	23 /*15	16 /*11	15 /*10	20 /*13	14 /*9	13 /*9	17 /*11	13 /*8	12 /*8
		3	14.0	Non-Exposed / *Exposed	61 /*47	61 /*33	52 /*30	61 /*41	50 /*30	43 /*27	61 /*37	42 /*27	38 /*24	56 /*33	38 /*24	34 /*22	49 /*30	34 /*22	31 /*20	43 /*27	31 /*20	28 /*18	36 /*23	26 /*17	24 /*15	30 /*19	22 /*14	20 /*13	26 /*17	19 /*12	17 /*11	22 /*14	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7
		4	18.4	Non-Exposed / *Exposed	60 /*46	60 /*33	51 /*30	60 /*41	49 /*29	42 /*27	60 /*36	42 /*26	38 /*24	55 /*33	37 /*24	34 /*22	48 /*30	34 /*22	31 /*20	43 /*27	31 /*20	28 /*18	35 /*23	26 /*17	24 /*15	30 /*19	22 /*14	20 /*13	26 /*17	19 /*12	17 /*11	22 /*14	16 /*11	15 /*10	20 /*13	14 /*9	13 /*8	17 /*11	13 /*8	12 /*7
		5	22.6	Non-Exposed / *Exposed	61	61 /*40	NA	61	61 /*36	NA	61 /*52	57 /*32	NA	61 /*43	47 /*29	NA	61 /*39	41 /*26	NA	61 /*35	37 /*24	NA	50 /*30	31 /*20	NA	39 /*25	26 /*17	NA	34 /*22	23 /*15	NA	29 /*19	20 /*13	NA	26 /*16	17 /*11	NA	23 /*15	15 /*10	NA
		6	26.6	Non-Exposed / *Exposed	63	63 /*40	NA	63 /*62	63 /*35	NA	63 /*51	55 /*32	NA	63 /*43	46 /*29	NA	63 /*38	41 /*26	NA	63 /*35	37 /*24	NA	48 /*29	31 /*20	NA	39 /*25	26 /*17	NA	33 /*21	23 /*15	NA	29 /*19	20 /*13	NA	25 /*16	17 /*11	NA	22 /*15	15 /*10	NA
		8	33.7	Non-Exposed / *Exposed	67 /*62	50 /*27	40 /*21	67 /*51	41 /*24	33 /*18	67 /*43	35 /*22	28 /*17	67 /*38	30 /*20	24 /*15	65 /*35	28 /*18	21 /*14	55 /*32	25 /*16	19 /*12	42 /*27	21 /*14	16 /*11	35 /*23	18 /*12	14 /*9	30 /*20	16 /*10	10 /*7	23 /*15	12 /*8	9 /*6	21 /*13	11 /*7	8 /*5	21 /*13	11 /*7	8 /*5
		10	39.8	Non-Exposed / *Exposed	67 /*59	47 /*27	39 /*20	67 /*49	39 /*24	33 /*18	67 /*42	34 /*21	28 /*16	67 /*38	30 /*19	24 /*15	61 /*34	27 /*18	21 /*14	53 /*31	25 /*16	19 /*12	41 /*26	21 /*14	16 /*10	35 /*23	18 /*12	14 /*9	30 /*20	15 /*10	12 /*8	26 /*17	14 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	11 /*7	8 /*5
		12	45.0	Non-Exposed / *Exposed	65 /*57	46 /*26	38 /*20	65 /*47	39 /*24	32 /*18	65 /*41	34 /*21	27 /*16	65 /*37	30 /*19	23 /*15	59 /*34	27 /*18	21 /*13	51 /*31	25 /*16	19 /*12	40 /*26	21 /*14	16 /*10	34 /*22	18 /*12	14 /*9	30 /*19	15 /*10	12 /*8	26 /*17	13 /*9	10 /*7	23 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5
	C	2	9.5	Non-Exposed / *Exposed	53 /*32	36 /*23	33 /*21	46 /*28	32 /*20	29 /*19	40 /*25	29 /*18	26 /*17	36 /*23	26 /*17	24 /*15	33 /*21	24 /*15	22 /*14	30 /*19	21 /*14	20 /*13	25 /*16	18 /*12	16 /*11	21 /*13	15 /*10	14 /*9	18 /*12	13 /*8	12 /*8	16 /*10	11 /*7	10 /*7	14 /*9	10 /*6	9 /*6	12 /*8	9 /*6	8 /*5
		3	14.0	Non-Exposed / *Exposed	52 /*31	36 /*23	33 /*21	45 /*28	32 /*20	29 /*19	40 /*25	29 /*18	26 /*17	36 /*23	26 /*17	24 /*15	32 /*21	23 /*15	21 /*14	29 /*19	21 /*14	19 /*13	25 /*16	18 /*11	16 /*11	21 /*13	15 /*10	14 /*9	18 /*12	13 /*8	12 /*8	16 /*10	11 /*7	10 /*7	14 /*9	10 /*6	9 /*6	12 /*8	9 /*6	8 /*5
		4	18.4	Non-Exposed / *Exposed	51 /*31	35 /*23	32 /*21	44 /*28	32 /*20	29 /*19	39 /*25	28 /*18	26 /*17	35 /*23	26 /*16	23 /*15	32 /*20	23 /*15	21 /*14	29 /*19	21 /*14	19 /*12	24 /*16	18 /*11	16 /*11	21 /*13	15 /*10	14 /*9	18 /*11	13 /*8	12 /*8	15 /*10	11 /*7	10 /*7	14 /*9	10 /*6	9 /*6	12 /*8	9 /*6	8 /*5
		5	22.6	Non-Exposed / *Exposed	61 /*41	44 /*27	NA	61 /*36	38 /*25	NA	60 /*33	34 /*22	NA	50 /*29	31 /*20	NA	42 /*27	28 /*18	NA	38 /*24	26 /*17	NA	32 /*20	22 /*14	NA	27 /*17	18 /*12	NA	23 /*15	16 /*10	NA	20 /*13	14 /*9	NA	18 /*11	12 /*8	NA	16 /*10	11 /*7	NA
		6	26.6	Non-Exposed / *Exposed	63 /*40	43 /*27	NA	63 /*36	38 /*24	NA	59 /*32	34 /*22	NA	49 /*29	31 /*20	NA	41 /*27	28 /*18	NA	38 /*24	25 /*16	NA	32 /*20	21 /*14	NA	27 /*17	18 /*12	NA	23 /*15	16 /*10	NA	20 /*13	14 /*9	NA	18 /*11	12 /*8	NA	16 /*10	11 /*7	NA
		8	33.7	Non-Exposed / *Exposed	67 /*37	29 /*19	23 /*14	58 /*33	26 /*17	20 /*13	49 /*30	23 /*15	18 /*12	41 /*27	21 /*14	16 /*10	38 /*24	19 /*12	15 /*10	34 /*22	18 /*11	13 /*9	29 /*19	15 /*10	11 /*7	25 /*16	13 /*8	10 /*6	21 /*14	11 /*7	8 /*5	18 /*12	9 /*6	7 /*5	16 /*11	8 /*5	6 /*4	14 /*9	7 /*5	6 /*4
		10	39.8	Non-Exposed / *Exposed	67 /*36	29 /*19	22 /*14	56 /*32	26 /*17	20 /*13	47 /*29	23 /*15	18 /*11	41 /*26	21 /*14	16 /*10	37 /*24	19 /*12	15 /*9	34 /*22	17 /*11	13 /*9	28 /*18	15 /*10	11 /*7	24 /*16	13 /*8	10 /*6	21 /*14	11 /*7	8 /*5	18 /*12	9 /*6	7 /*5	16 /*10	8 /*5	6 /*4	14 /*9	7 /*5	6 /*4
		12	45.0	Non-Exposed / *Exposed	64 /*35	28 /*18	22 /*14	54 /*32	25 /*17	19 /*13	45 /*29	23 /*15	18 /*11	40 /*26	21 /*14	16 /*10	36 /*24	19 /*12	14 /*9	33 /*22	17 /*11	13 /*9	28 /*18	15 /*9	11 /*7	24 /*16	12 /*8	10 /*6	21 /*14	11 /*7	8 /*5	18 /*12	9 /*6	7 /*5	16 /*10	8 /*5	6 /*4	14 /*9	7 /*5	6 /*4
D	2	9.5	Non-Exposed / *Exposed	42 /*26	30 /*19	27 /*17	37 /*23	27 /*17	24 /*16	33 /*21	24 /*15	22 /*14	30 /*19	22 /*14	20 /*13	27 /*17	20 /*13	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4	
	3	14.0	Non-Exposed / *Exposed	41 /*26	30 /*19	27 /*17	37 /*23	26 /*17	24 /*16	33 /*21	24 /*15	22 /*14	29 /*19	21 /*14	20 /*13	27 /*17	19 /*12	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4	
	4	18.4	Non-Exposed / *Exposed	41 /*26	29 /*19	27 /*17	36 /*23	26 /*17	24 /*15	32 /*21	24 /*15	22 /*14	29 /*19	21 /*14	20 /*13	26 /*17	19 /*12	18 /*11	24 /*16	18 /*11	16 /*10	20 /*13	15 /*10	14 /*9	17 /*11	13 /*8	12 /*7	15 /*10	11 /*7	10 /*6	13 /*8	9 /*6	9 /*6	11 /*7	8 /*5	8 /*5	10 /*6	7 /*5	7 /*4	
	5	22.6	Non-Exposed / *Exposed	61 /*34	36 /*23	NA	52 /*30	32 /*20	NA	43 /*27	29 /*18	NA	38 /*24	26 /*17	NA	35 /*22	23 /*15	NA	32 /*20	21 /*14	NA	26 /*17	18 /*12	NA	23 /*15	15 /*10	NA	19 /*13	13 /*9	NA	17 /*11	11 /*7	NA	15 /*10	10 /*6	NA	13 /*8	9 /*6	NA	
	6	26.6	Non-Exposed / *Exposed	63 /*34	35 /*23	NA	51 /*30	32 /*20	NA	42 /*27	28 /*18	NA	38 /*24	26 /*17	NA	34 /*22	23 /*15	NA	31 /*20	21 /*14	NA	26 /*17	18 /*12	NA	22 /*14	15 /*10	NA	19 /*12	13 /*8	NA	17 /*11	11 /*7	NA	15 /*10	10 /*6	NA	13 /*8	9 /*6	NA	
	8	33.7	Non-Exposed / *Exposed	52 /*31	24 /*16	18 /*12	43 /*27	22 /*14	17 /*11	38 /*25	19 /*13	15 /*10	34 /*22	18 /*11	13 /*9	31 /*20	16 /*10	12 /*8	29 /*19	15 /*9	11 /*7	24 /*16	12 /*8	9 /*6	21 /*13	11 /*7	8 /*5	18 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	14 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3	
	10	39.8	Non-Exposed / *Exposed	50 /*30	24 /*16	18 /*12	42 /*27	21 /*14	16 /*11	37 /*24	19 /*13	15 /*10	34 /*22	17 /*11	13 /*9	31 /*20	16 /*10	12 /*8	28 /*18	15 /*9	11 /*7	24 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4	5 /*3	
	12	45.0	Non-Exposed / *Exposed	48 /*30	24 /*15	18 /*12	41 /*27	21 /*14	16 /*11	37 /*24	19 /*12	15 /*10	33 /*22	17 /*11	13 /*9	30 /*20	16 /*10	12 /*8	28 /*18	14 /*9	11 /*7	24 /*15	12 /*8	9 /*6	20 /*13	10 /*7	8 /*5	18 /*11	9 /*6	7 /*4	15 /*10	8 /*5	6 /*4	13 /*9	7 /*4	5 /*3	12 /*8	6 /*4		

Roof Tech - Mini Gen II - ASCE 7-16 Hip - Flush-Mount - 30 Foot Max Mean Roof Height - Plywood Only - Portrait 60 Cell - 60 mm Long Screws																																								
Ground Snow Load (psf)	Exposure Category	Roof Pitch X:12	Roof Angle (Degrees)	Wind Speed (mph) ---->	95			100			105			110			115			120			130			140			150			160			170			180		
				Roof Area ---->	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3			
40	B	2	9.5	Non-Exposed / *Exposed	45	45/*30	45/*31	45/*41	45/*30	44/*27	45/*37	43/*27	39/*24	45/*33	38/*24	35/*22	45/*30	34/*22	31/*20	44/*27	31/*20	29/*18	36/*23	26/*17	24/*15	31/*19	22/*14	20/*13	26/*17	19/*12	17/*11	23/*15	16/*11	15/*10	20/*13	14/*9	13/*9	17/*11	13/*8	12/*8
		3	14.0	Non-Exposed / *Exposed	41	41/*33	41/*30	41	41/*30	41/*27	41/*37	41/*27	38/*24	41/*33	38/*24	34/*22	41/*30	34/*22	31/*20	41/*27	31/*20	28/*18	36/*23	26/*17	24/*15	30/*19	22/*14	20/*13	26/*17	19/*12	17/*11	22/*14	16/*11	15/*10	20/*13	14/*9	13/*8	17/*11	13/*8	12/*7
		4	18.4	Non-Exposed / *Exposed	40	40/*33	40/*30	40	40/*29	40/*27	40/*36	40/*26	38/*24	40/*33	37/*24	34/*22	40/*30	34/*22	31/*20	40/*27	31/*20	28/*18	35/*23	26/*17	24/*15	30/*19	22/*14	20/*13	26/*17	19/*12	17/*11	22/*14	16/*11	15/*10	20/*13	14/*9	13/*8	17/*11	13/*8	12/*7
		5	22.6	Non-Exposed / *Exposed	41	41/*40	NA	41	41/*36	NA	41	41/*36	NA	41	41/*32	NA	41	41/*39	41/*26	NA	41	41/*35	37/*24	NA	41	39/*25	26/*17	NA	34/*22	23/*15	NA	29/*19	20/*13	NA	26/*16	17/*11	NA	23/*15	15/*10	NA
		6	26.6	Non-Exposed / *Exposed	42	42/*40	NA	42	42/*35	NA	42	42/*35	NA	42	42/*32	NA	42	42/*38	41/*26	NA	42	42/*35	37/*24	NA	42	39/*25	26/*17	NA	33/*21	23/*15	NA	29/*19	20/*13	NA	25/*16	17/*11	NA	22/*15	15/*10	NA
		8	33.7	Non-Exposed / *Exposed	48	48/*27	40/*21	48	48/*24	33/*18	48/*43	28/*17	48/*38	30/*20	24/*15	48/*35	28/*18	21/*14	48/*32	25/*16	19/*12	42/*27	21/*14	16/*11	35/*23	18/*12	14/*9	30/*20	16/*10	12/*8	27/*17	14/*9	10/*7	23/*15	12/*8	9/*6	21/*13	11/*7	8/*5	
		10	39.8	Non-Exposed / *Exposed	55	47/*27	39/*20	55/*49	39/*24	33/*18	55/*42	34/*21	28/*16	55/*38	30/*19	24/*15	55/*34	27/*18	21/*14	53/*31	25/*16	19/*12	41/*26	21/*14	16/*10	35/*23	18/*12	14/*9	30/*20	15/*10	12/*8	26/*17	14/*9	10/*7	23/*15	12/*8	9/*6	20/*13	11/*7	8/*5
		12	45.0	Non-Exposed / *Exposed	64/*57	46/*26	38/*20	64/*47	39/*24	32/*18	64/*41	34/*21	27/*16	64/*37	30/*19	23/*15	59/*34	27/*18	21/*13	51/*31	25/*16	19/*12	40/*26	21/*14	16/*10	34/*22	18/*12	14/*9	30/*19	15/*10	12/*8	26/*17	13/*9	10/*7	23/*15	12/*8	9/*6	20/*13	10/*7	8/*5
	C	2	9.5	Non-Exposed / *Exposed	45/*32	36/*23	33/*21	45/*28	32/*20	29/*19	40/*25	29/*18	26/*17	36/*23	26/*17	24/*15	33/*21	24/*15	22/*14	30/*19	21/*14	20/*13	25/*16	18/*12	16/*11	21/*13	15/*10	14/*9	18/*12	13/*8	12/*8	16/*10	11/*7	10/*7	14/*9	10/*6	9/*6	12/*8	9/*6	8/*5
		3	14.0	Non-Exposed / *Exposed	41/*31	36/*23	33/*21	41/*28	32/*20	29/*19	40/*25	29/*18	26/*17	36/*23	26/*17	24/*15	32/*21	23/*15	21/*14	29/*19	21/*14	19/*13	25/*16	18/*11	16/*11	21/*13	15/*10	14/*9	18/*12	13/*8	12/*8	16/*10	11/*7	10/*7	14/*9	10/*6	9/*6	12/*8	9/*6	8/*5
		4	18.4	Non-Exposed / *Exposed	40/*31	35/*23	32/*21	40/*28	32/*20	29/*19	39/*25	28/*18	26/*17	35/*23	26/*16	23/*15	32/*20	23/*15	21/*14	29/*19	21/*14	19/*12	24/*16	18/*11	16/*11	21/*13	15/*10	14/*9	18/*11	13/*8	12/*8	15/*10	11/*7	10/*7	14/*9	10/*6	9/*6	12/*8	9/*6	8/*5
		5	22.6	Non-Exposed / *Exposed	41	41/*27	NA	41/*36	38/*25	NA	41/*33	34/*22	NA	41/*29	31/*20	NA	41/*27	28/*18	NA	38/*24	26/*17	NA	32/*20	22/*14	NA	27/*17	18/*12	NA	23/*15	16/*10	NA	20/*13	14/*9	NA	18/*11	12/*8	NA	16/*10	11/*7	NA
		6	26.6	Non-Exposed / *Exposed	42/*40	42/*27	NA	42/*36	38/*24	NA	42/*32	34/*22	NA	42/*29	31/*20	NA	41/*27	28/*18	NA	38/*24	25/*16	NA	32/*20	21/*14	NA	27/*17	18/*12	NA	23/*15	16/*10	NA	20/*13	14/*9	NA	18/*11	12/*8	NA	16/*10	11/*7	NA
		8	33.7	Non-Exposed / *Exposed	48/*37	29/*19	23/*14	48/*33	26/*17	20/*13	48/*30	23/*15	18/*12	48/*31	21/*14	16/*10	38/*24	19/*12	15/*10	34/*22	18/*11	13/*9	29/*19	21/*14	11/*7	25/*16	13/*8	10/*6	21/*14	11/*7	8/*5	18/*12	9/*6	7/*5	16/*11	8/*5	6/*4	14/*9	7/*5	6/*4
		10	39.8	Non-Exposed / *Exposed	55/*36	29/*19	22/*14	55/*32	26/*17	20/*13	47/*29	23/*15	18/*11	41/*26	21/*14	16/*10	37/*24	19/*12	15/*9	34/*22	17/*11	13/*9	28/*18	15/*10	11/*7	24/*16	13/*8	10/*6	21/*14	11/*7	8/*5	18/*12	9/*6	7/*5	16/*10	8/*5	6/*4	14/*9	7/*5	6/*4
		12	45.0	Non-Exposed / *Exposed	64/*35	28/*18	22/*14	54/*32	25/*17	19/*13	45/*29	23/*15	18/*11	40/*26	21/*14	16/*10	36/*24	19/*12	14/*9	33/*22	17/*11	13/*9	28/*18	15/*9	11/*7	24/*16	12/*8	10/*6	21/*14	11/*7	8/*5	18/*12	9/*6	7/*5	16/*10	8/*5	6/*4	14/*9	7/*5	6/*4
	D	2	9.5	Non-Exposed / *Exposed	42/*26	30/*19	27/*17	37/*23	27/*17	24/*16	33/*21	24/*15	22/*14	30/*19	22/*14	20/*13	27/*17	20/*13	18/*11	24/*16	18/*11	16/*10	20/*13	15/*10	14/*9	17/*11	13/*8	12/*7	15/*10	11/*7	10/*6	13/*8	9/*6	9/*6	11/*7	8/*5	8/*5	10/*6	7/*5	7/*4
		3	14.0	Non-Exposed / *Exposed	41/*26	30/*19	27/*17	37/*23	26/*17	24/*16	33/*21	24/*15	22/*14	29/*19	21/*14	20/*13	27/*17	19/*12	18/*11	24/*16	18/*11	16/*10	20/*13	15/*10	14/*9	17/*11	13/*8	12/*7	15/*10	11/*7	10/*6	13/*8	9/*6	9/*6	11/*7	8/*5	8/*5	10/*6	7/*5	7/*4
		4	18.4	Non-Exposed / *Exposed	40/*26	29/*19	27/*17	36/*23	26/*17	24/*15	32/*21	24/*15	22/*14	29/*19	21/*14	20/*13	26/*17	19/*12	18/*11	24/*16	18/*11	16/*10	20/*13	15/*10	14/*9	17/*11	13/*8	12/*7	15/*10	11/*7	10/*6	13/*8	9/*6	9/*6	11/*7	8/*5	8/*5	10/*6	7/*5	7/*4
		5	22.6	Non-Exposed / *Exposed	41/*34	36/*23	NA	41/*30	32/*20	NA	41/*27	29/*18	NA	38/*24	26/*17	NA	35/*22	23/*15	NA	32/*20	21/*14	NA	26/*17	18/*12	NA	23/*15	15/*10	NA	19/*13	13/*9	NA	17/*11	11/*7	NA	15/*10	10/*6	NA	13/*8	9/*6	NA
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		8	33.7	Non-Exposed / *Exposed	48/*31	24/*16	18/*12	43/*27	22/*14	17/*11	38/*25	19/*13	15/*10	34/*22	18/*11	13/*9	31/*20	16/*10	12/*8	29/*19	15/*9	11/*7	24/*16	12/*8	9/*6	21/*13	11/*7	8/*5	18/*11	9/*6	7/*4	15/*10	8/*5	6/*4	14/*9	7/*4	5/*3	12/*8	6/*4	5/*3
		10	39.8	Non-Exposed / *Exposed	50/*30	24/*16	18/*12	42/*27	21/*14	16/*11	37/*24	19/*13	15/*10	34/*22	17/*11	13/*9	31/*20	16/*10	12/*8	28/*18	15/*9	11/*7	24/*15	12/*8	9/*6	20/*13	10/*7	8/*5	18/*11	9/*6	7/*4	15/*10	8/*5	6/*4	13/*9	7/*4	5/*3	12/*8	6/*4	5/*3
		12	45.0	Non-Exposed / *Exposed	48/*30	24/*15	18/*12	41/*27	21/*14	16/*11	37/*24	19/*12	15/*10	33/*22	17/*11	13/*9	30/*20	16/*10	12/*8	28/*18	14/*9	11/*7	24/*15	12/*8	9/*6	20/*13	10/*7	8/*5	18/*11	9/*6	7/*4	15/*10	8/*5	6/*4	13/*9	7/*4	5/*3	12/*8	6/*4	5/*3
50	B	2	9.5	Non-Exposed / *Exposed																																				

Note: See SML "Roof Area" conversion chart and associated ASCE 7-16 Roof Zones on pages 6 - 12.

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Table notes for all tables – tabulated values are based on the following criteria:

1. ASCE 7-16.
2. Building mean roof height (MRH) = 15 or 30 ft maximum, see specific tables for each height limit.
3. Risk Category = II.
4. Solar panel tributary (trib.) widths used to determine the tributary area and maximum spacing = 40 inches for Landscape tables, 68 inches for Portrait 60 Cell tables & 77.5 inches for Portrait 72 Cell tables.
 - a. Please note that PV panels with larger trib. widths may be used, eg. 90 cell panels, provided they exceed the tributary width for the given table used for installation.
 - b. It is acceptable to decrease the maximum span shown in the tables for larger tributary widths by the following equation: “New Max. Span” = “Table Max. Span” x “Table Trib. Width” / “New Trib. Width”; eg. If “New Trib. Width” = 44 inches, you can use the Landscape Tables (40 inch Trib. Width), but you will need to reduce the span as follows: “New Max. Span” = “Table Max. Span” x “40 inches” / “44 inches”. So if the “Table Max. Span” = 96 inches the “New Max. Span” = “96 inches” x “40 inches” / “44 inches” = “87 inches”.
5. Solar panel & rail dead load = approximately 4.0 psf.
6. γ_E = edge array factor, has been taken as 1.0 for Non-Exposed Panels and 1.5 for *Exposed Panels as defined in ASCE 7-16 section 29.4.4. If the tables show two (2) mount spacings (eg. 85 / *52) the second lower mount spacing with the * is for *Exposed Panels (eq. *52). If the tables only show one (1) mount spacing then the same mount spacing is applicable for both Non-Exposed and *Exposed Panels.
7. γ_a = solar panel pressure equalization factor, has been reduced in two ways. Per SEAOC PV2-2017 section 5.3.5, and ASCE 7-16 Figure 29.4-8 for the actual trib area to each mount. The following limits have been used to determine the SEAOC reduction: for Rafter installations the minimum average gap between panels is 0.625” with a maximum top of panel height of 7”, and for OSB and Plywood Only installations the minimum average gap between panels is 0.5” with a maximum top of panel height of 5”, see Exhibits ‘A1’ & ‘A2’ for panel height dimension illustrations. Alternatively, for OSB and Plywood Only installations the average gap between panels may be 0.7” with a maximum top of panel height of 7”.
8. OSB shall be 24/16 APA rated sheathing minimum (7/16” thick).
9. Plywood shall be 32/16 APA rated sheathing minimum (15/32” thick).
10. Sheathing shall be free of defects including, but not limited to water damage and delamination and must be evaluated by the project Engineer of Record (EOR).
11. Roof rafters or trusses spaced at 24” on center maximum, and must be evaluated by the project EOR for structural integrity and capacity as required by the governing jurisdiction.
12. PV panels must be supported per the manufacturer’s required orientation, location and/or spacing. Verification of PV module capacity to support the loads associated with the given array shall be the responsibility of the Contractor or Owner and not Roof Tech or Starling Madison Lofquist, Inc.
13. The mounts may be installed through a maximum of 2 layers of composite asphalt roof shingles or maximum 20 gage metal decking provided the metal is predrilled as wood screws are not designed to penetrate the metal decking.

Our analysis assumes that the connections and associated hardware are installed in a workmanlike manner in accordance with the RT-Mini II Installation Manual and generally accepted standards of construction practice. It is the responsibility of the project EOR to verify that the strength of the roof framing meets the minimum properties used in the tests and can safely support the maximum imposed loads stated within this document. Starling Madison Lofquist, Inc. and Roof Tech assume no liability beyond what is specifically shown in this letter. Additional information is available at the Roof Tech web site, <http://www.roof-tech.us/>.

Please feel free to contact me at your convenience if you have any questions.

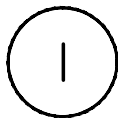
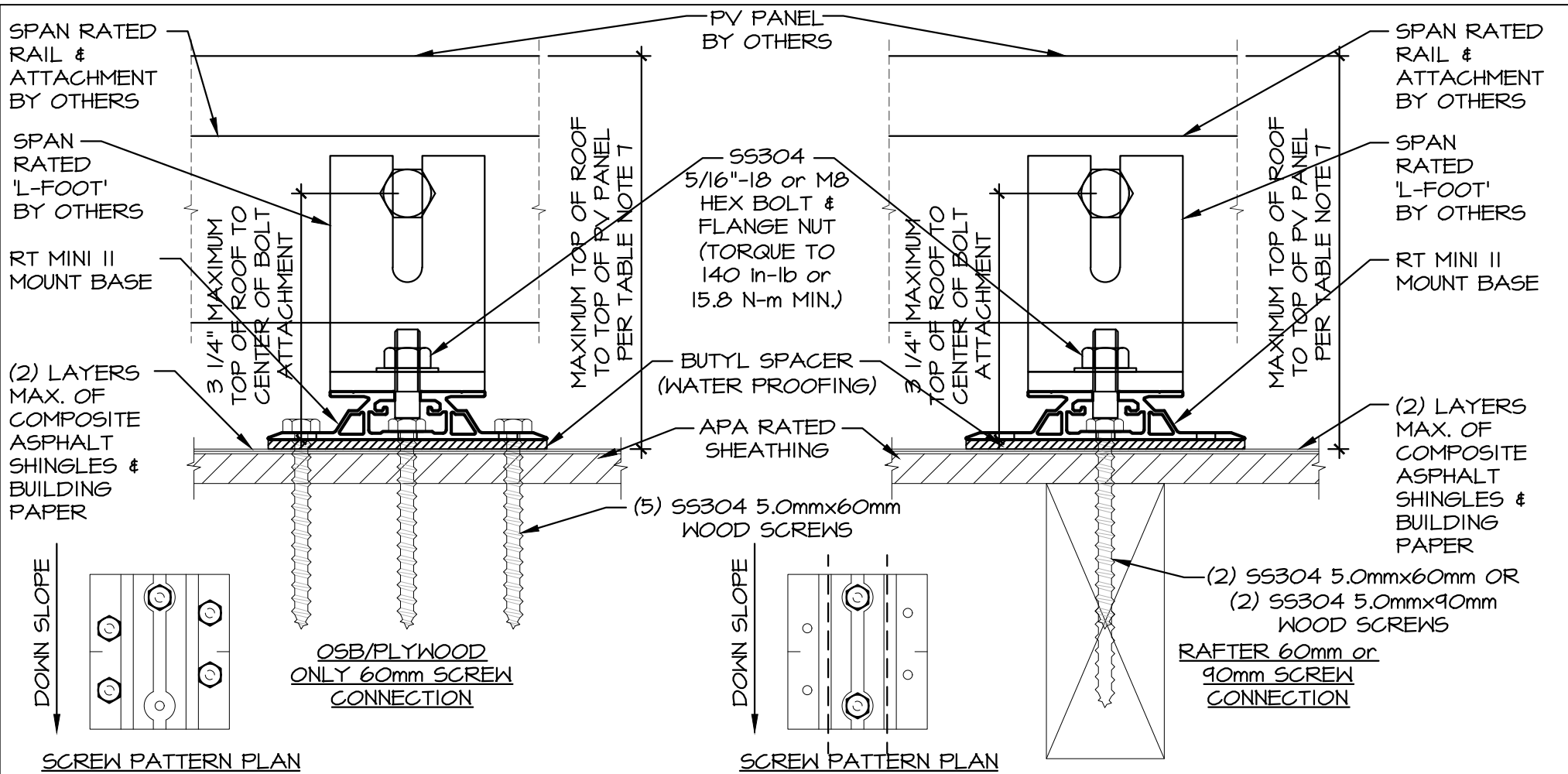
Respectfully yours,



Jesse Light, S.E., P.E.
Principal (VP) / Sr. Structural Engineer

EXPIRES: 11/30/2026
FIRM REG. NO.: F 6845-452-2

Rusmir Begic, P.E., M.S.E.
Associate / Licensed Professional Engineer



RAIL AND 'L-FOOT' ORIENTATION

SCALE: N.T.S.



Consulting Structural and
Forensic Engineers

Starling Madison Lofquist, Inc.

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Phoenix, Arizona 85034

(602) 438-2500

fax. (602) 438-2505

DRAWN BY: J.S.

DESIGNED BY: J.L.

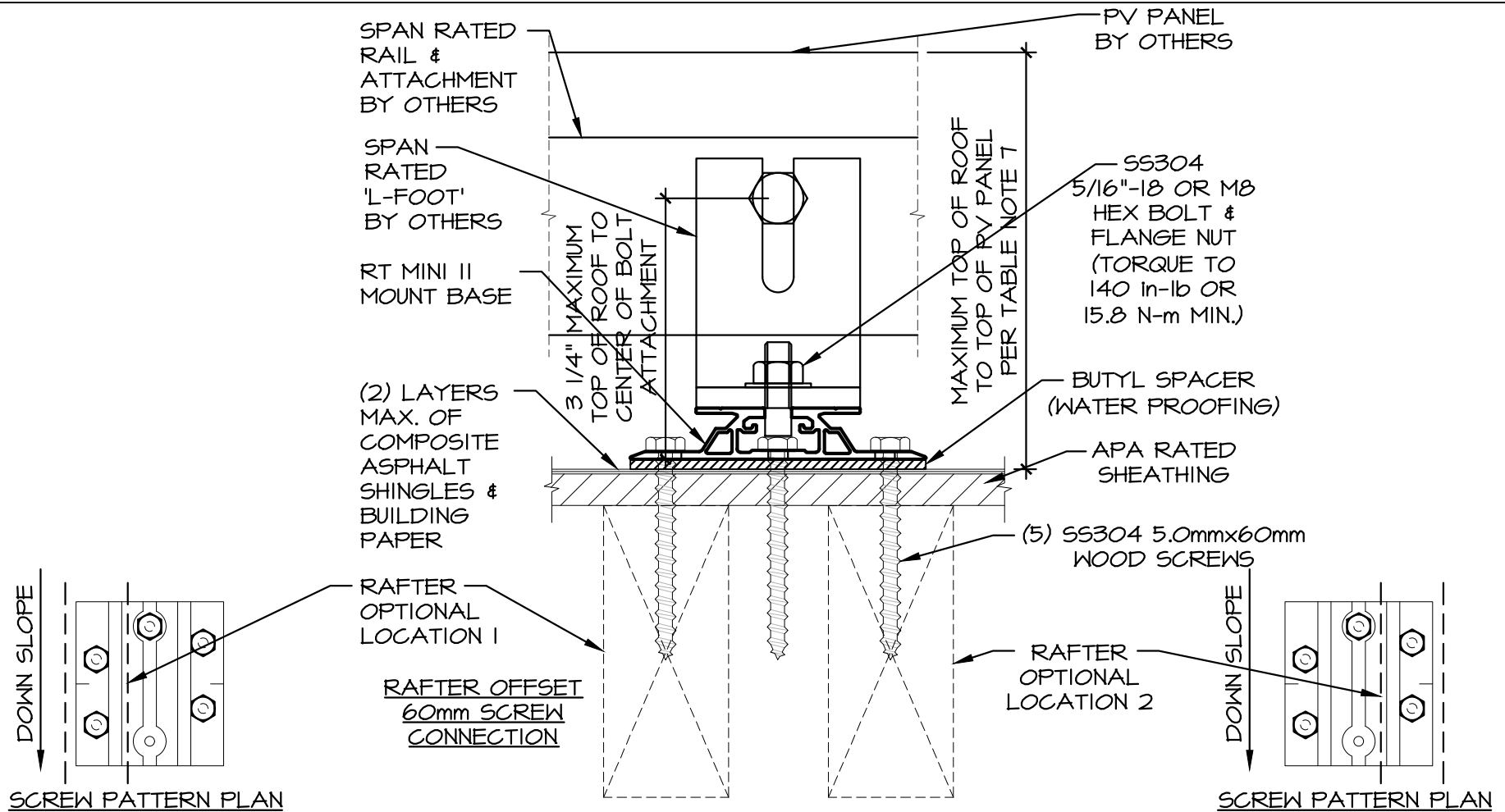
DATE: 3-7-23

JOB NO: 471-22

ROOF TECH
RT-MINI II MOUNT
WITH L-FOOT, RAILS, &
PV PANELS BY OTHERS

ROOF TECH, INC.
10620 TREENA ST., SUITE 230
SAN DIEGO, CA 92131

EXHIBIT A1



2 RAIL AND 'L-FOOT' ORIENTATION

SCALE: N.T.S.



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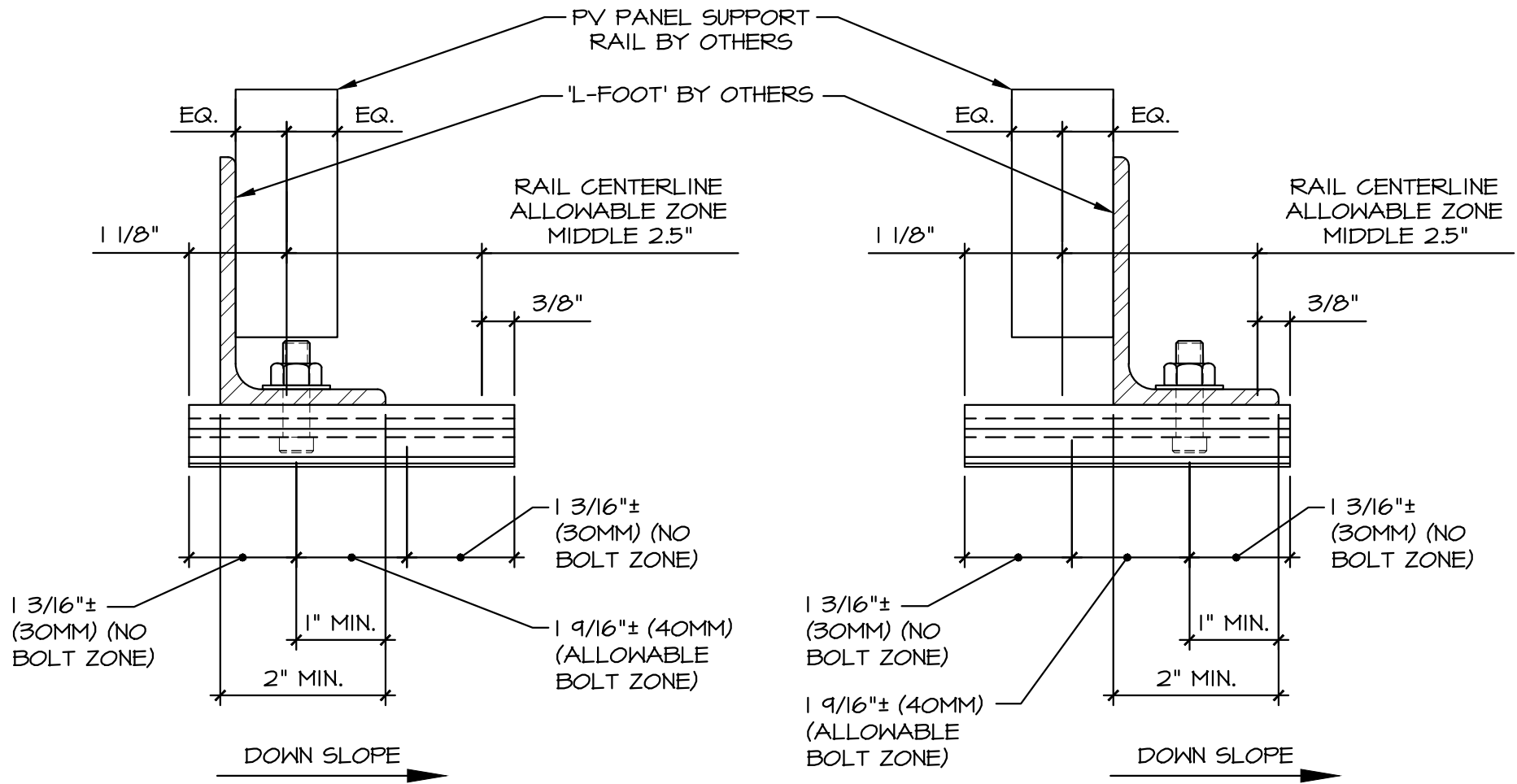
DRAWN BY: J.S.
DESIGNED BY: J.L.

DATE: 3-7-23
JOB NO: 471-22

ROOF TECH
RT-MINI II MOUNT
WITH L-FOOT, RAILS, &
PV PANELS BY OTHERS

ROOF TECH, INC.
10620 TREENA ST., SUITE 230
SAN DIEGO, CA 92131

EXHIBIT A2



3

RAIL AND 'L-FOOT' ORIENTATION

SCALE: N.T.S.



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DESIGNED BY: J.L.

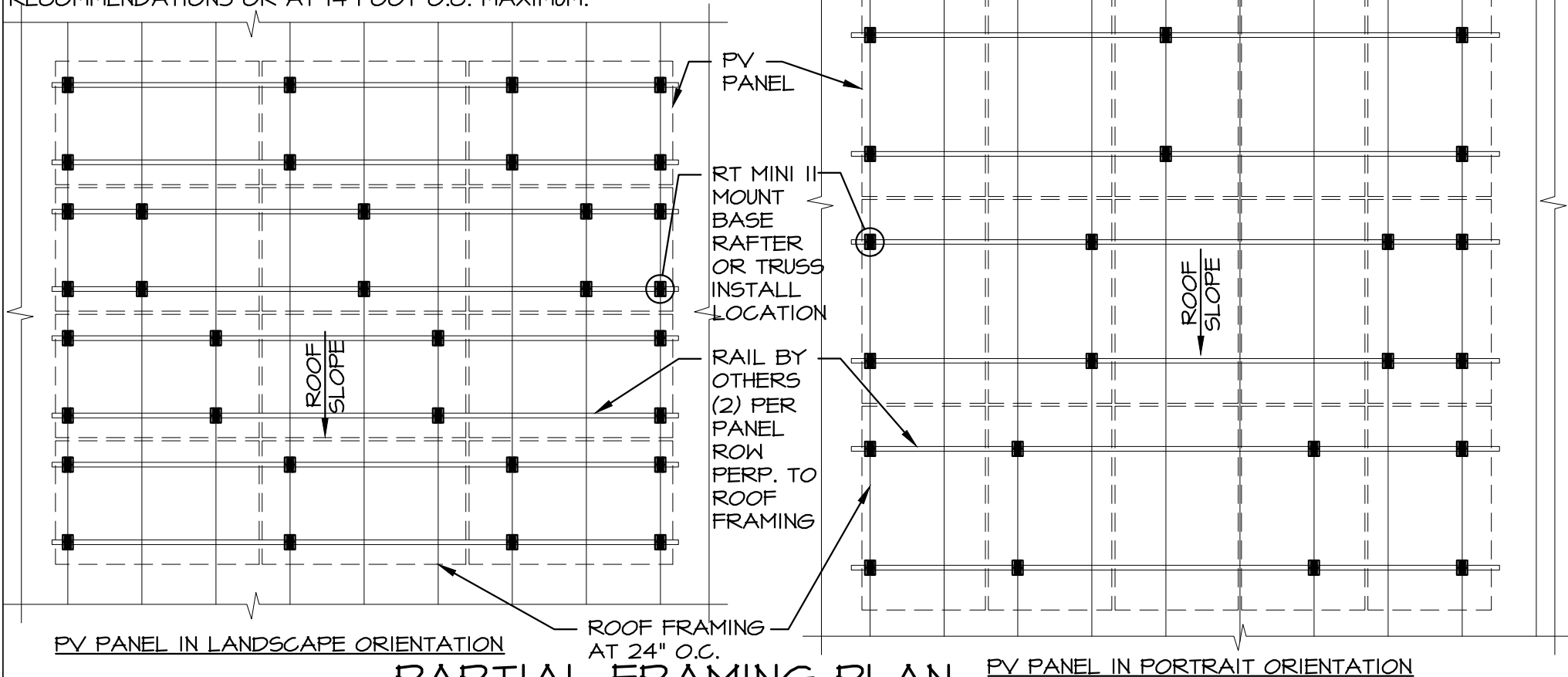
DATE: 3-7-23
JOB NO: 471-22

ROOF TECH
RT-MINI II MOUNT
WITH L-FOOT, RAILS, &
PV PANELS BY OTHERS

ROOF TECH, INC.
10620 TREENA ST., SUITE 230
SAN DIEGO, CA 92131

EXHIBIT A3

NOTE: RT MINI II MOUNT BASE TO BE ORIENTED WITH LONG DIRECTION DOWN ROOF SLOPE AS SHOWN. SKIP LOAD ROOF WHERE POSSIBLE TO EVENLY DISTRIBUTE LOAD TO ROOF FRAMING. THERMAL EXPANSION RAIL SPLICES AT MANUFACTURER'S RECOMMENDATIONS OR AT 14 FOOT O.C. MAXIMUM.



4

PARTIAL FRAMING PLAN (RAFTER INSTALLATION)

SCALE: N.T.S.



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DESIGNED BY: J.L.

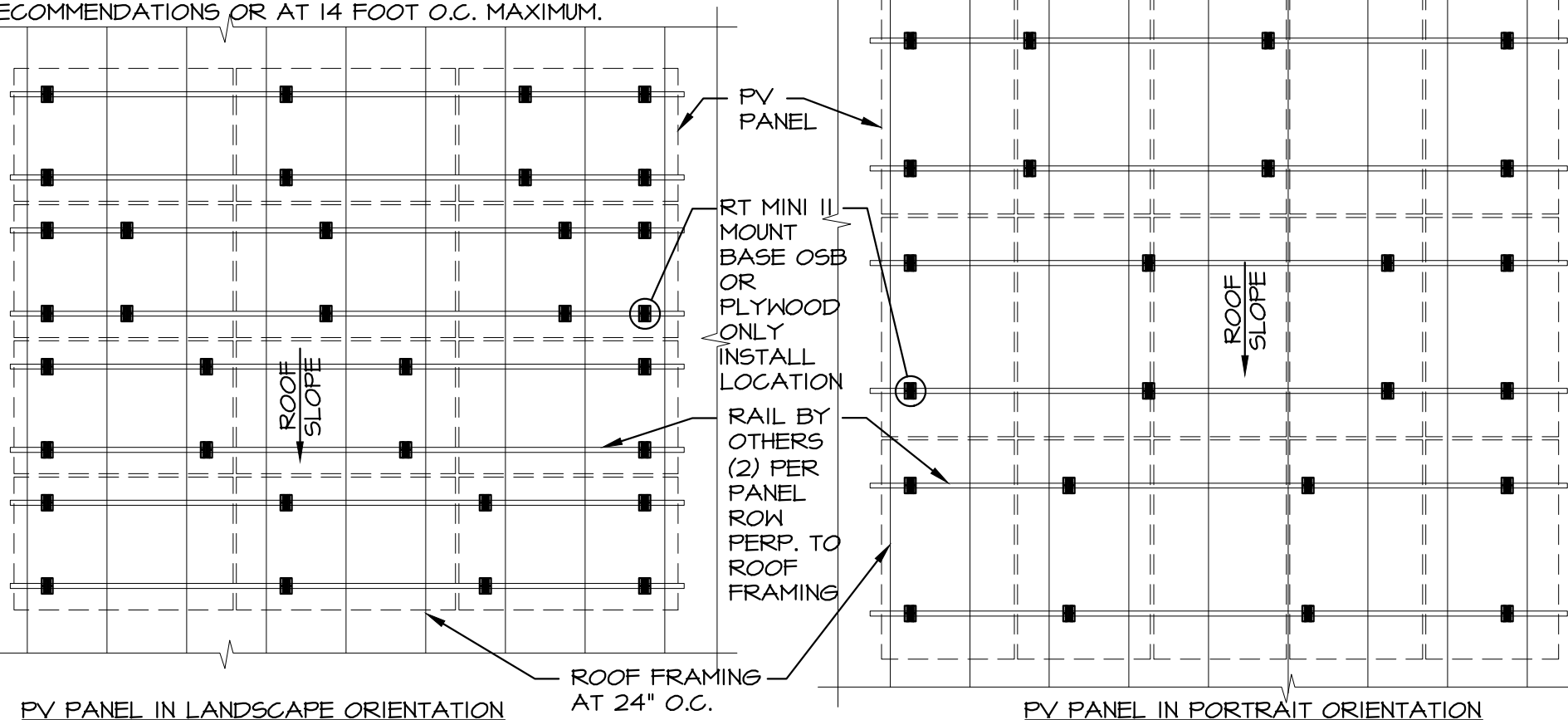
DATE: 3-7-23
JOB NO: 471-22

ROOF TECH
RT-MINI II MOUNT
WITH L-FOOT, RAILS, &
PV PANELS BY OTHERS

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SAN DIEGO, CA 92131

EXHIBIT A4

NOTE: RT MINI II MOUNT BASE TO BE ORIENTED WITH LONG DIRECTION DOWN ROOF SLOPE AS SHOWN. SKIP LOAD ROOF WHERE POSSIBLE TO EVENLY DISTRIBUTE LOAD TO ROOF FRAMING. THERMAL EXPANSION RAIL SPLICES AT MANUFACTURER'S RECOMMENDATIONS OR AT 14 FOOT O.C. MAXIMUM.



PARTIAL FRAMING PLAN (OSB/PLYWOOD ONLY INSTALLATION)

5

SCALE: N.T.S.



Consulting Structural and
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DRAWN BY: J.S.

DESIGNED BY: J.L.

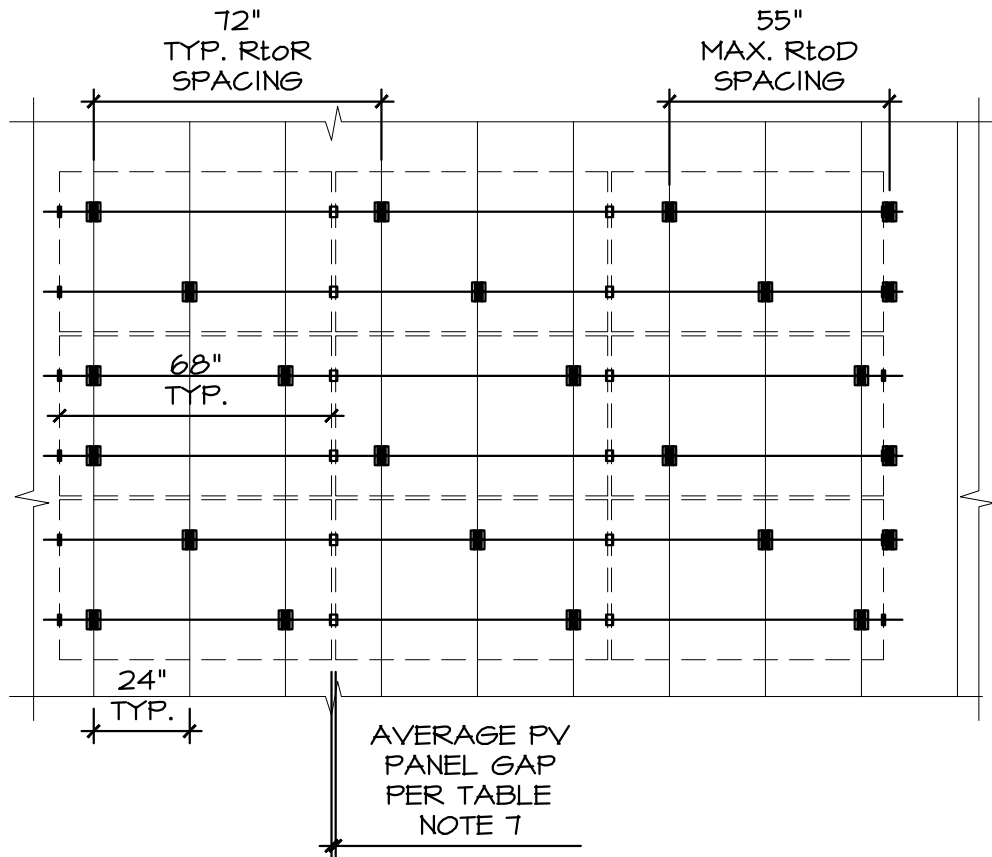
DATE: 3-7-23

JOB NO: 471-22

ROOF TECH
RT-MINI II MOUNT
WITH L-FOOT, RAILS, &
PV PANELS BY OTHERS

ROOF TECH, INC.
10620 TREENA ST., SUITE 230
SAN DIEGO, CA 92131

EXHIBIT A5



LANDSCAPE EXAMPLE:

DESIGN CRITERIA:

- ASCE 7-16 GABLE ROOF
 - 30 ft MEAN ROOF HEIGHT
 - 80 PSF GROUND SNOW LOAD
 - 140 MPH WIND SPEED
 - 4:12 ROOF PITCH (18.4 DEGREE ROOF ANGLE)
 - WIND EXPOSURE 'C'
 - ROOF ZONE 2; NON-EXPOSED PV PANELS
 - 60 OR 72 CELL PV PANELS (60 CELL SHOWN)
 - RAFTER AND PLYWOOD DECKING ONLY
- COMBINED INSTALLATION 60mm SCREWS
NOTE: SIMILAR PROCEDURE FOR OSB DECKING ONLY

RAFTER-TO-RAFTER (RtoR) = 92 in (72 in ACTUAL)
DECKING-TO-DECKING (DtoD) = 17 in

$RtoD = \text{RAFTER-TO-DECKING} = [(RtoR) + (DtoD)] / 2$

$RtoD = [92\text{in} + 17\text{in}] / 2 = 55\text{in MAX.}$

PLEASE NOTE: CHECK WITH PV PANEL MANUFACTURER FOR ALLOWABLE SUPPORT LOCATIONS FOR THE PROJECT SPECIFIC DESIGN LOADS. SUPPORT LOCATIONS SHOWN MAY NOT BE ALLOWED FOR ALL PV PANELS.

PARTIAL FRAMING PLAN AND EXAMPLE CALCULATION (COMBINED RAFTER-TO-DECKING ONLY LANDSCAPE INSTALL)

SCALE: 1/4" = 1'-0"



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EXHIBIT B1

